

SUBJECT: Data Visualization with Power BI and Tableau

Practical 4: Designing Effective and Visually Appealing Dashboards

AIM: Performing Data Analysis using Calculations in Tableau

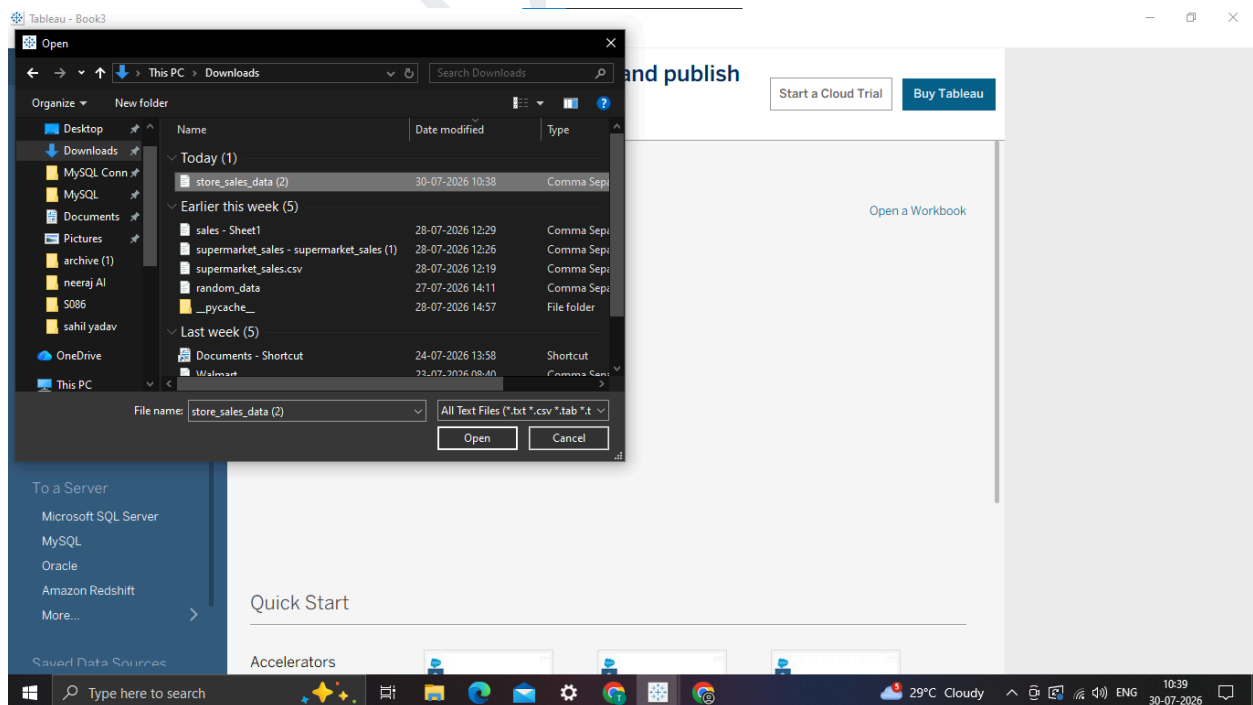
- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

Dataset: Indian Store Data

A: Creating Basic Calculated Fields Using Formulas

1. Connecting to the Dataset

1. Launch Tableau. Under the Connect panel, click on Text File, select your store_sales_data.csv, and click Open.



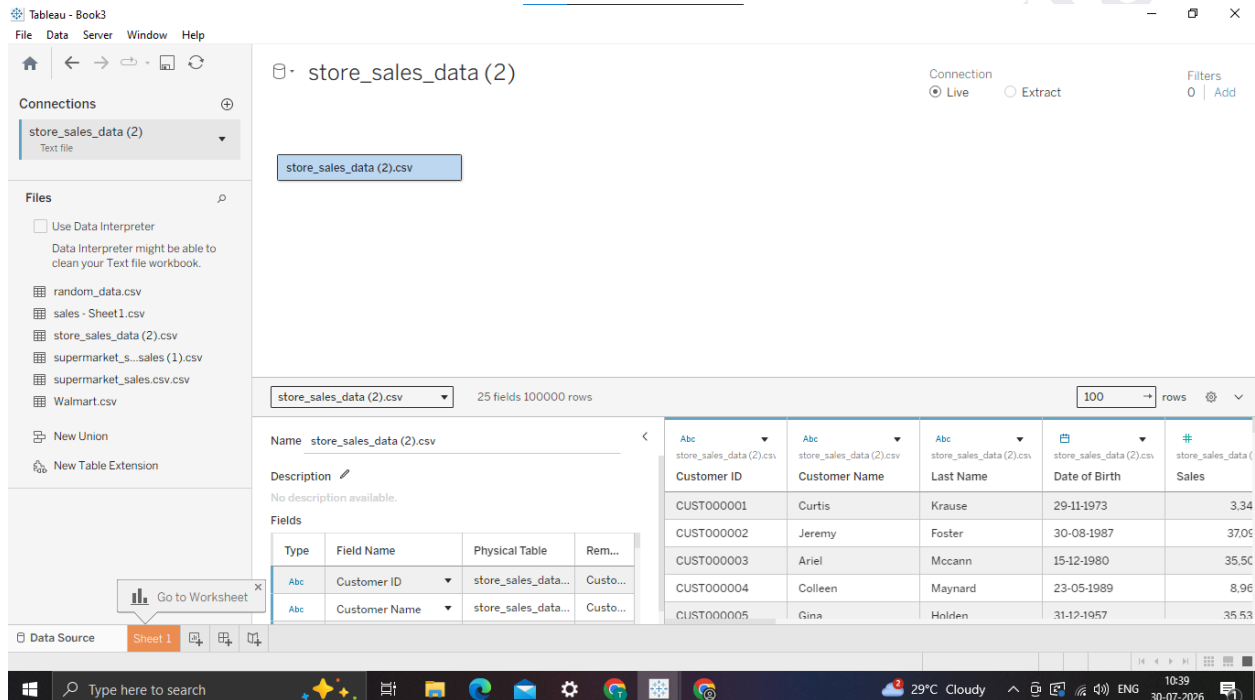
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2. Click on Sheet 1 at the bottom-left corner to enter your workspace.

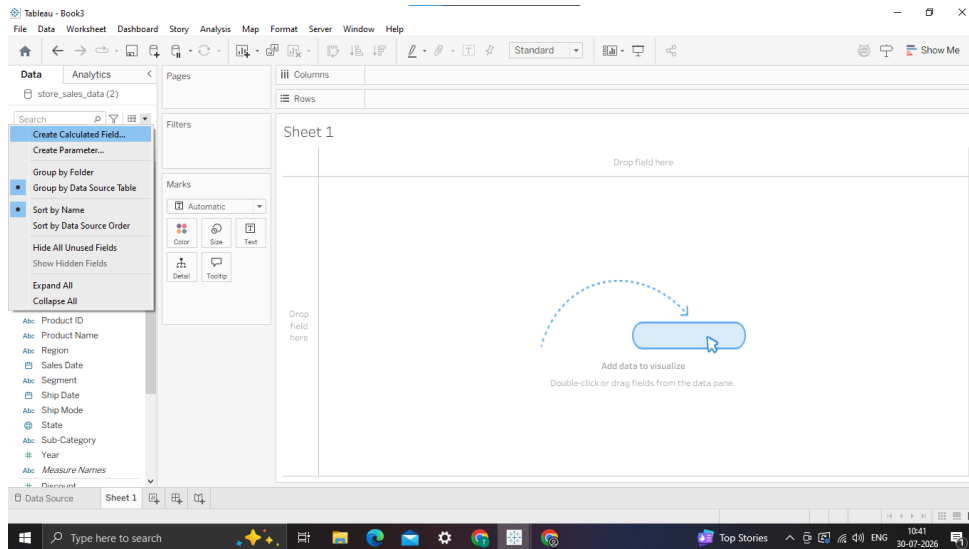


2. Building a Basic Custom Calculation

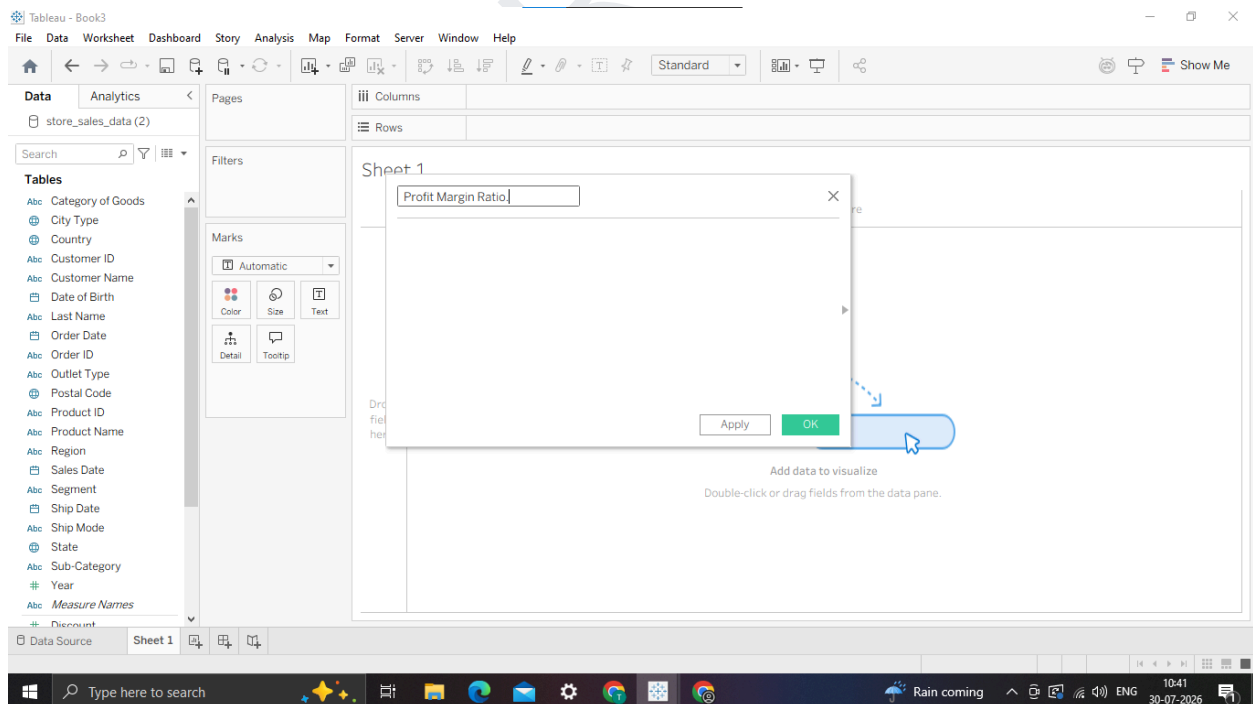
1. Look at your left-hand Data Pane. Right-click anywhere in the blank space of the pane (or click the small downward triangle next to the search bar) and select Create Calculated Field...

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2. In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.



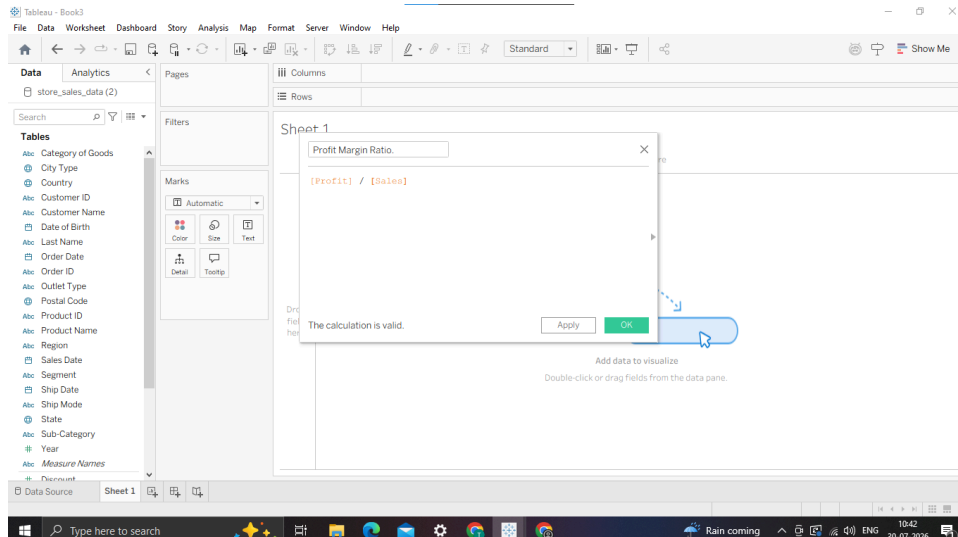
3. In the main workspace box, enter the following basic mathematical formula:

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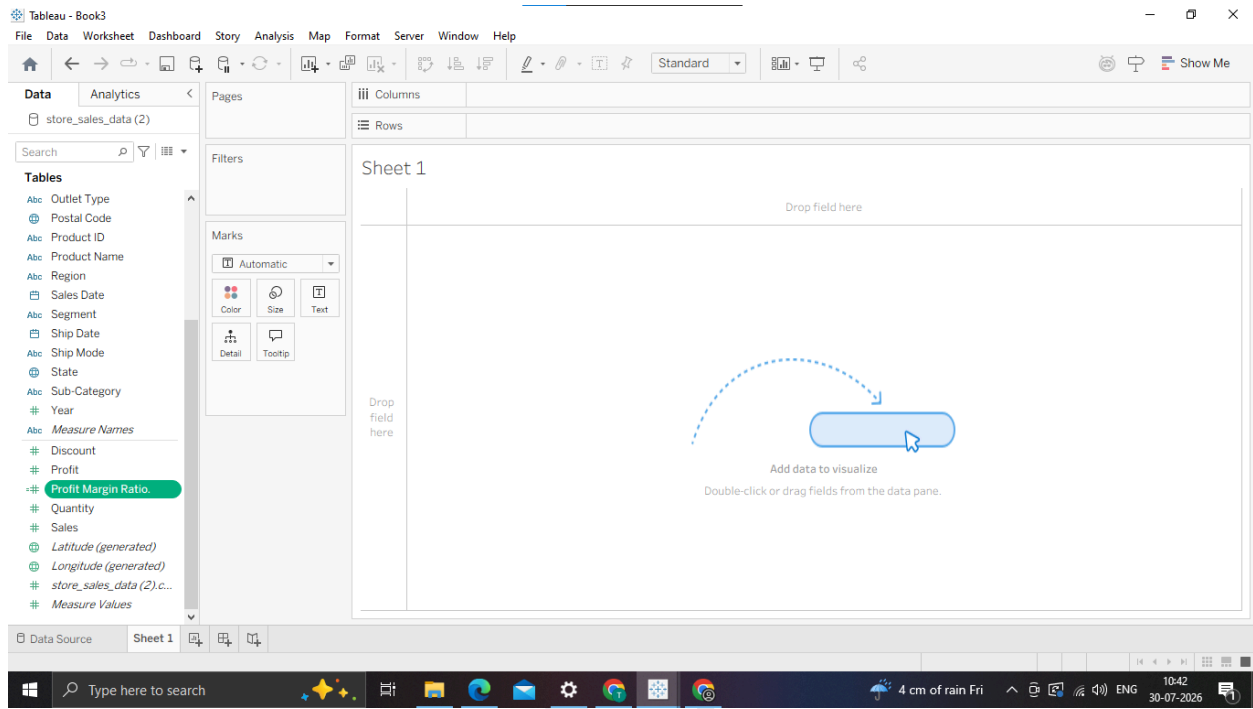
[Profit] / [Sales]



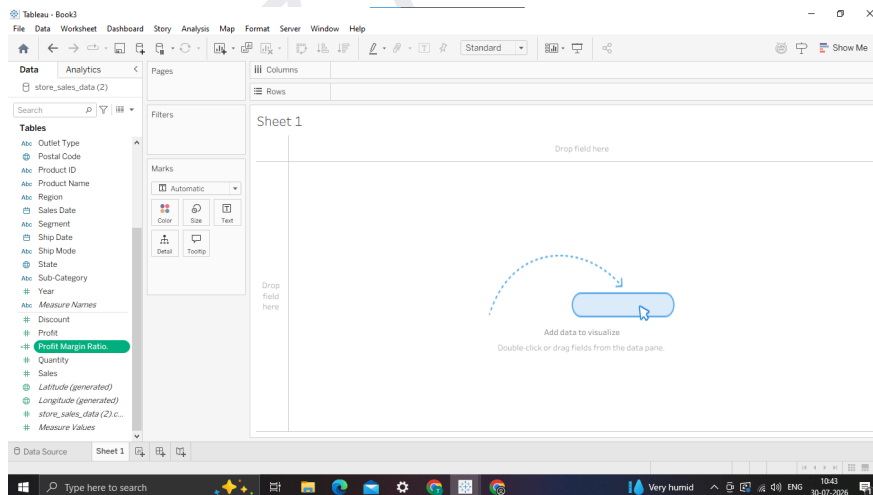
4. Look at the bottom-left corner of the editor box. It should say "The calculation is valid."
Click OK.

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5. Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.



2. Putting the Field to Use

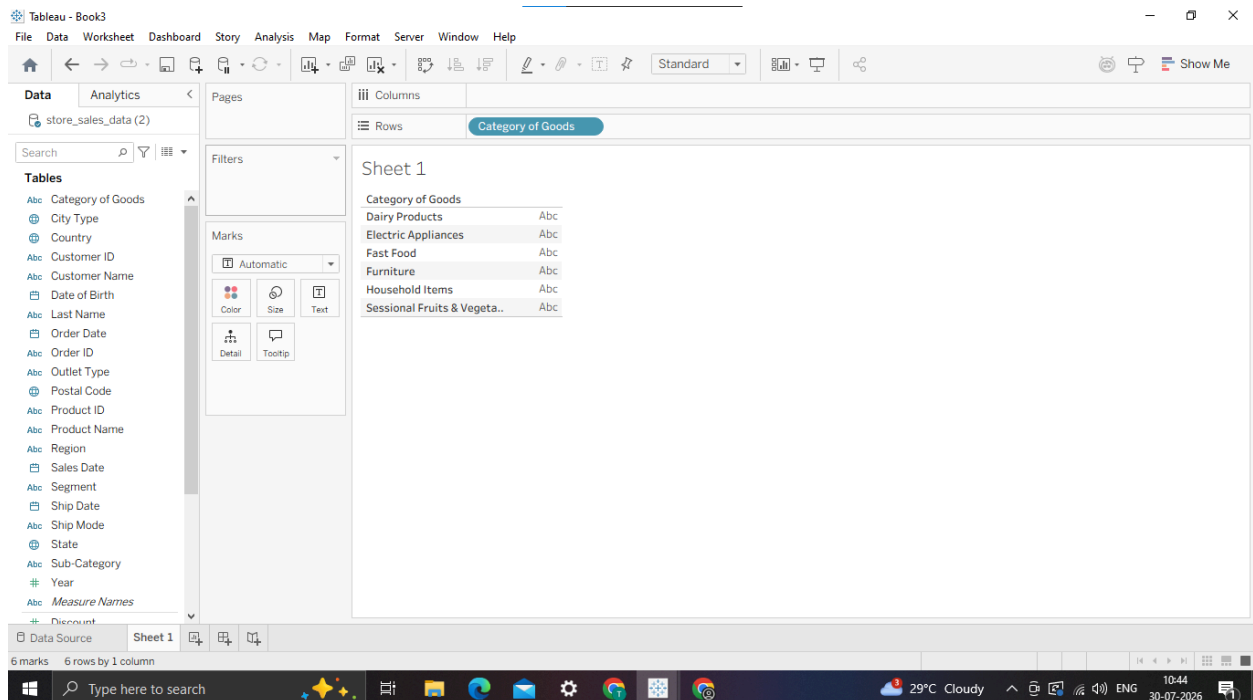
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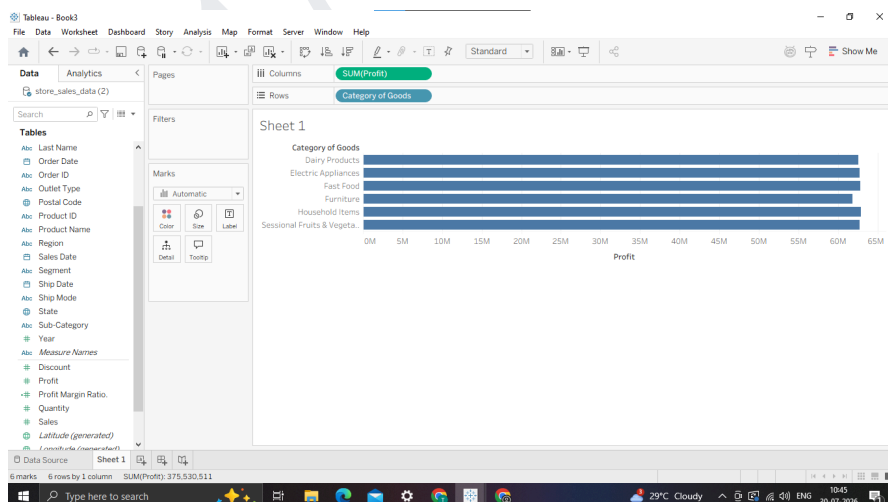
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1. Drag the Category of Goods dimension from your Data Pane and drop it onto the Rows shelf.



2. Drag your newly created Profit Margin Ratio calculated field and drop it onto the Columns shelf.

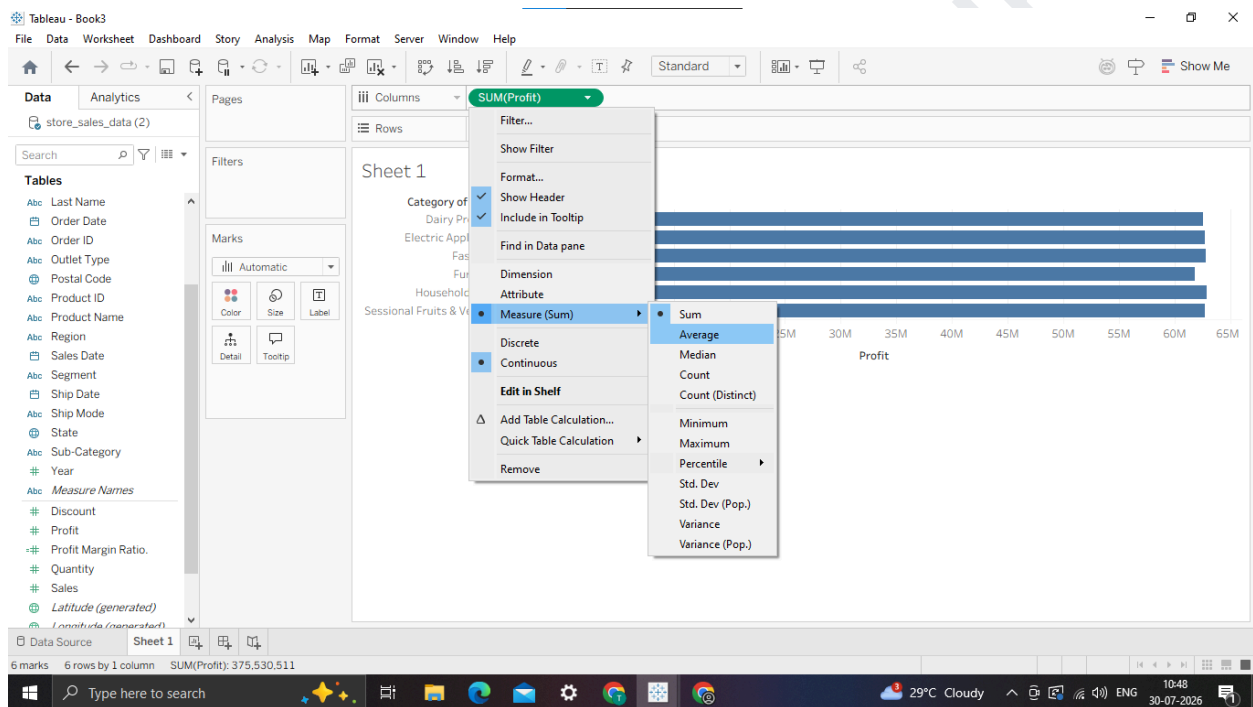


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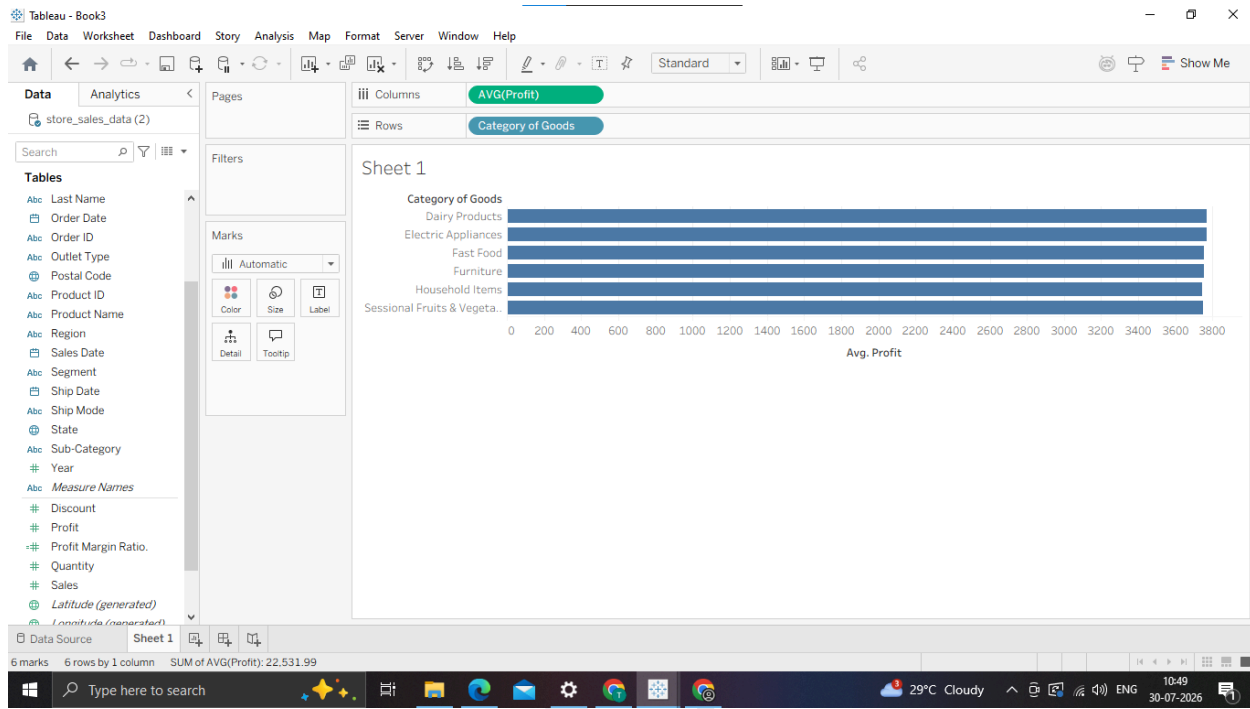
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3. Right-click the Profit Margin Ratio pill now sitting on the Columns shelf, hover over Measure (Sum), and change it to Average to see the true margin distribution across product types.



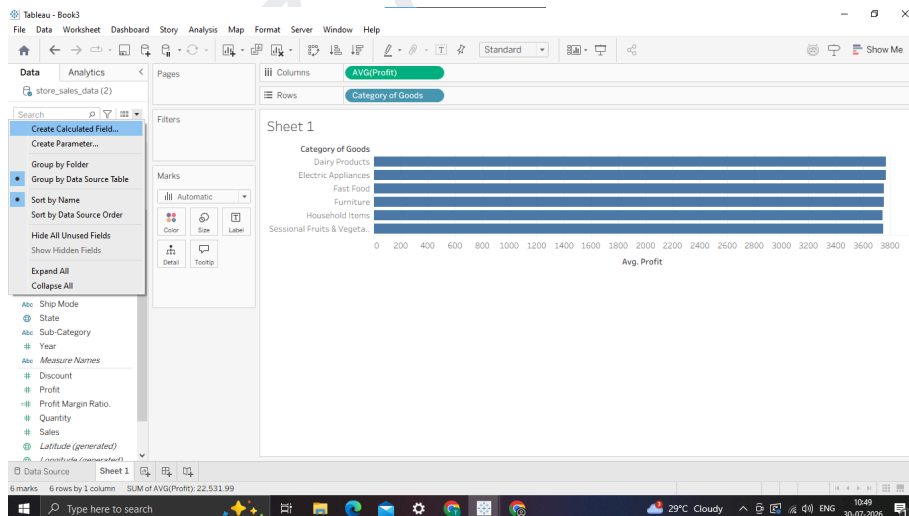
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B: Performing Row-Level Calculations

1. Creating a Row-Level Calculated Field

1. Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...



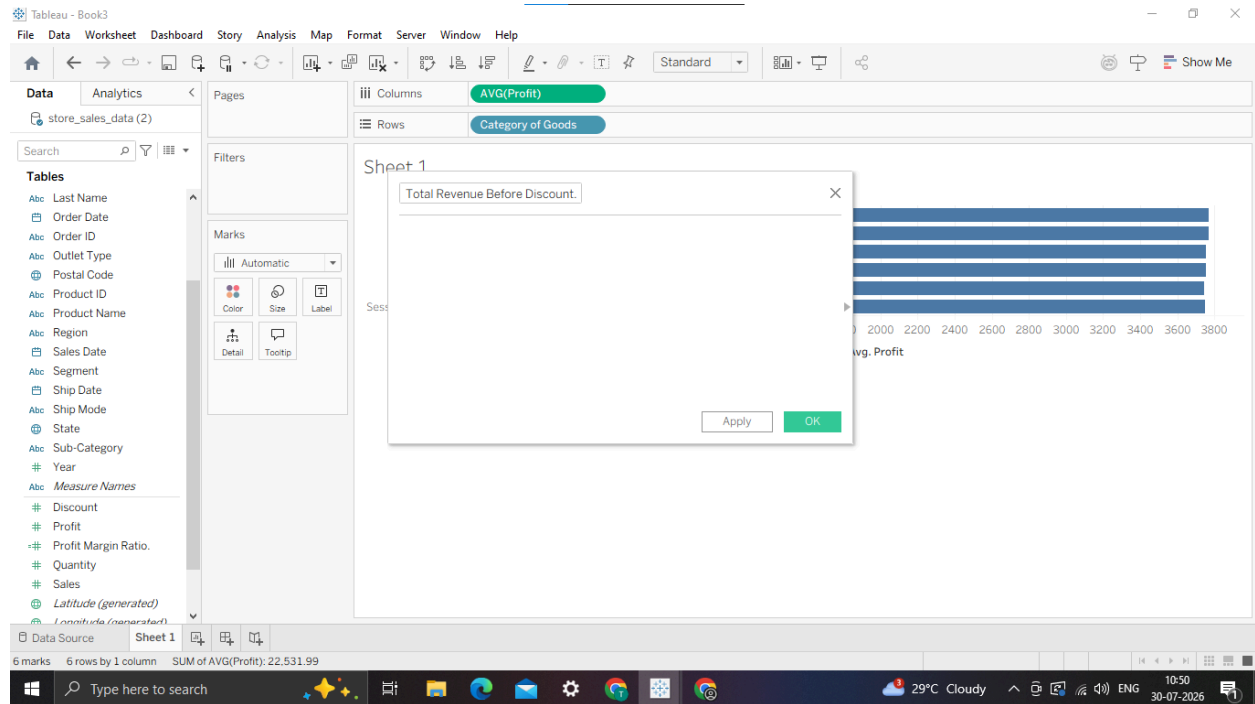
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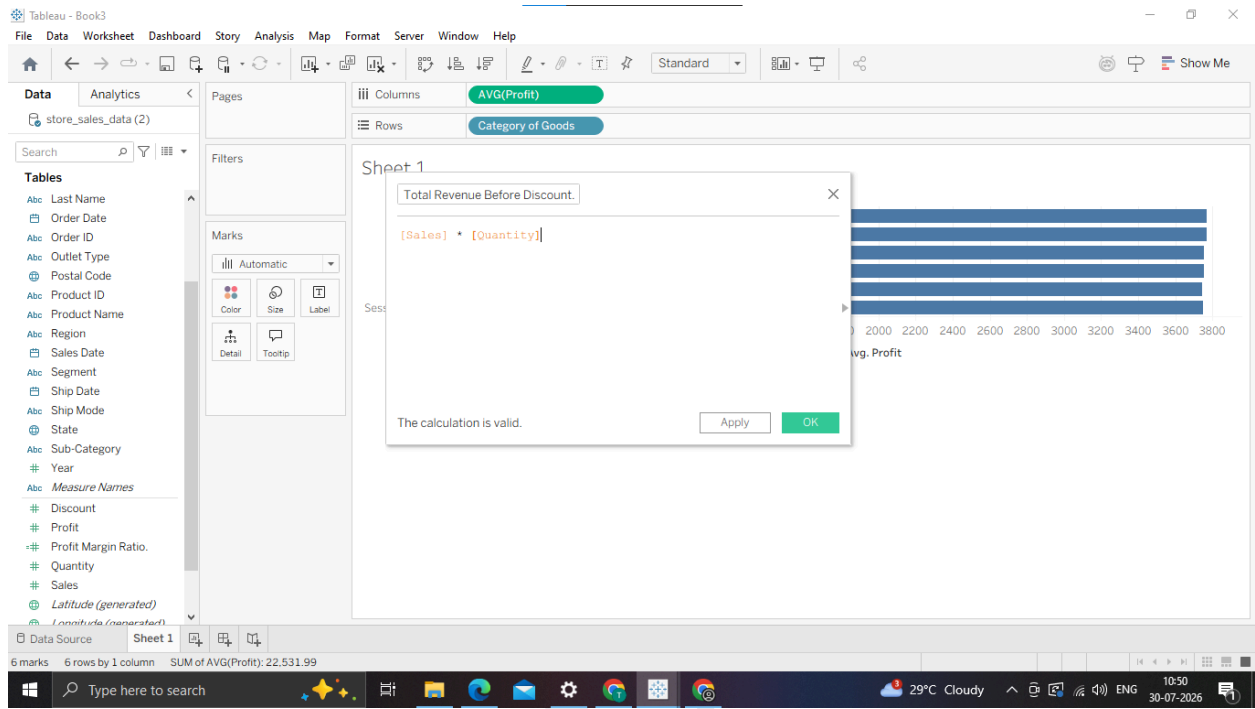
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2. Name the field: Total Revenue Before Discount.

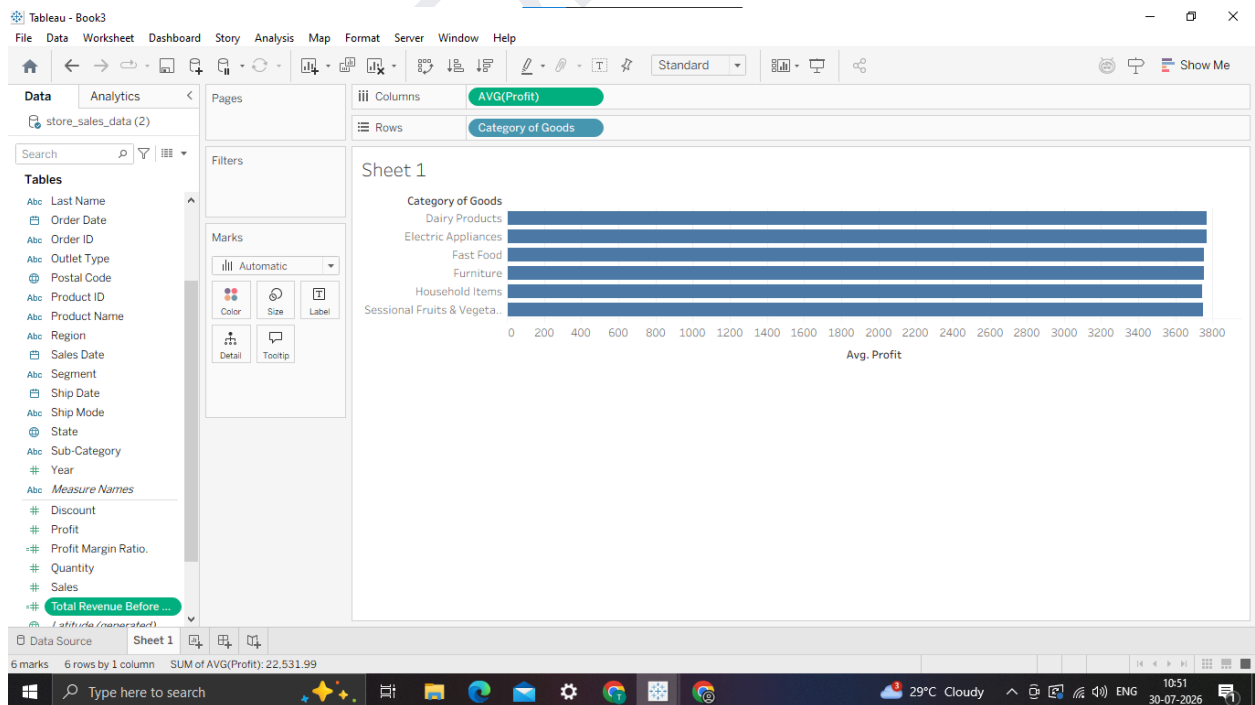


3. In the calculation formula canvas area, input the following expression:

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4. Verify the message at the bottom reads "The calculation is valid" and click OK.

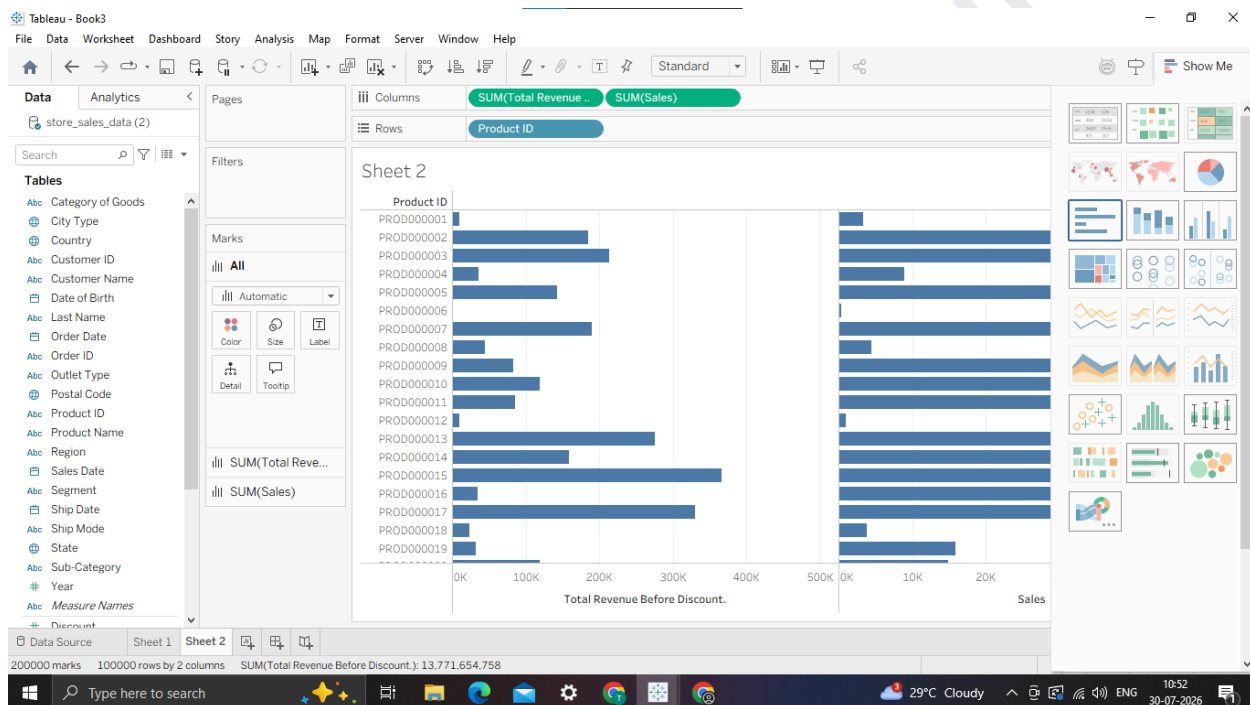


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2. Putting the Field to Use

1. Drag the Product Name dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Total Revenue Before Discount calculated field and drop it onto the Columns shelf.
3. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.



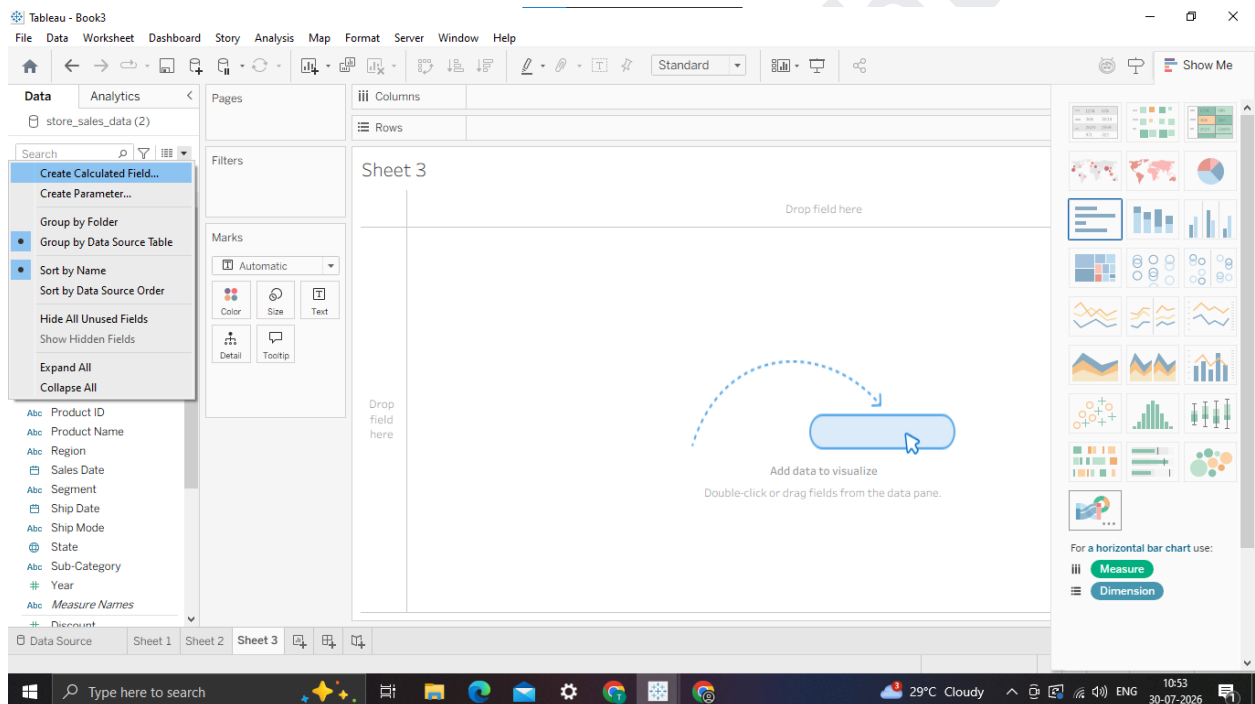
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C: Performing Aggregate-Level Calculations

1. Creating an Aggregate Calculated Field

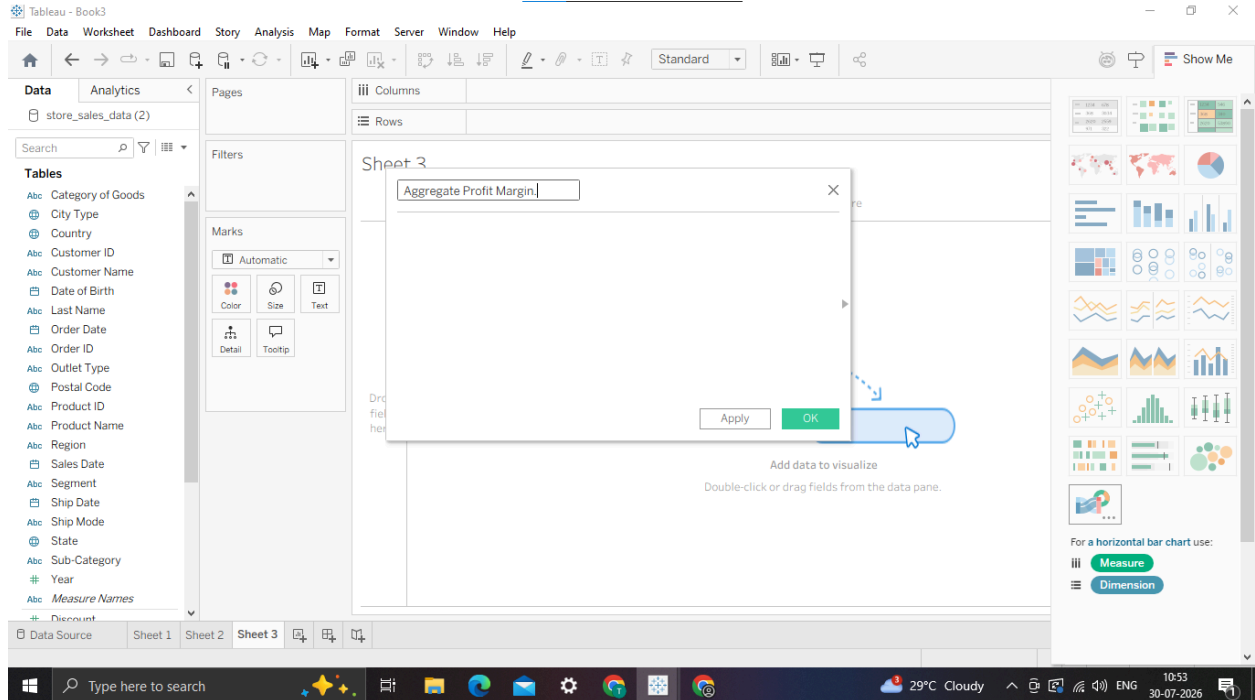
1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



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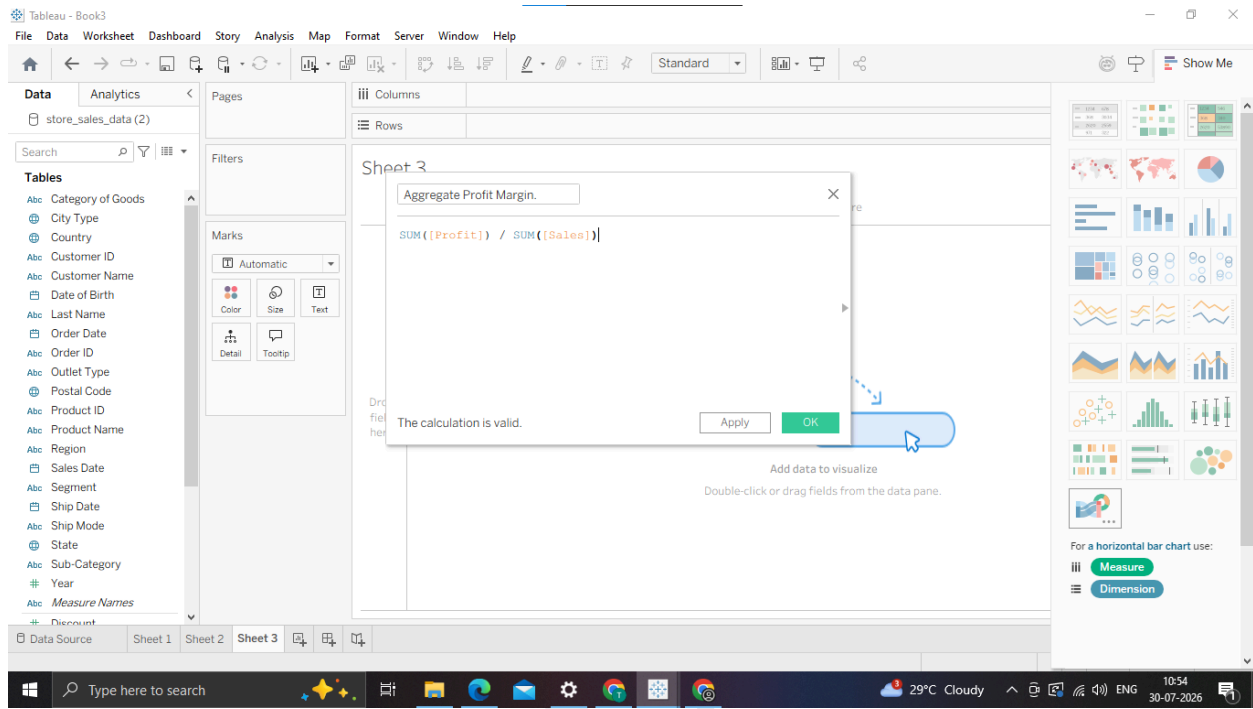
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2. Name the field:

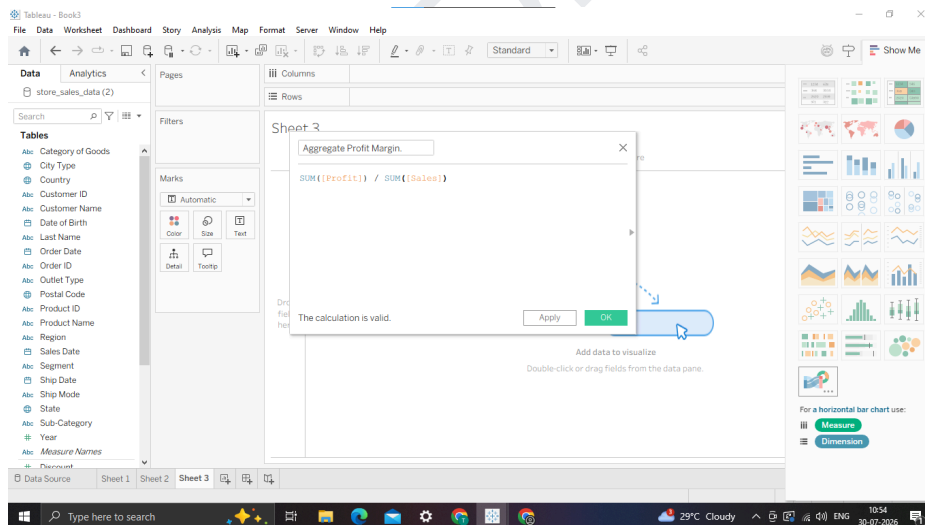


3. In the calculation formula window, type the following expression:
$$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$$

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4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



2. Putting the Field to Use

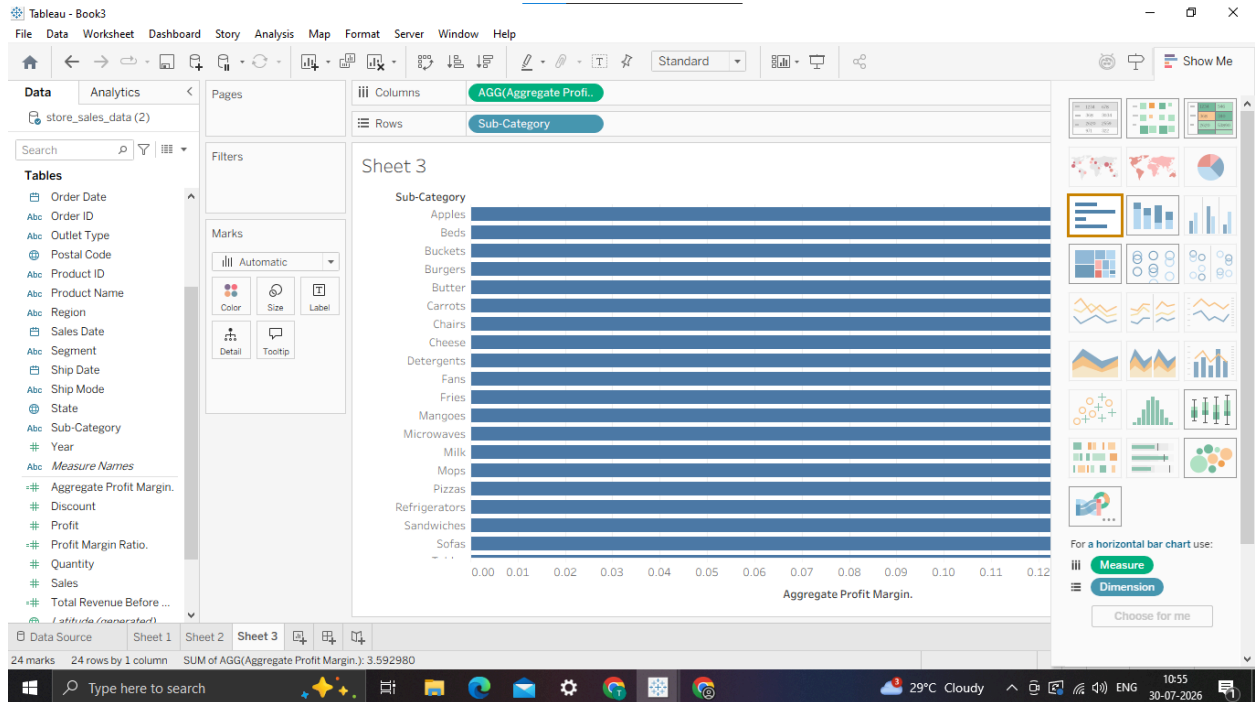
1. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.

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3. Observe closely: Look at the pill on the Columns shelf. It displays as AGG(Aggregate

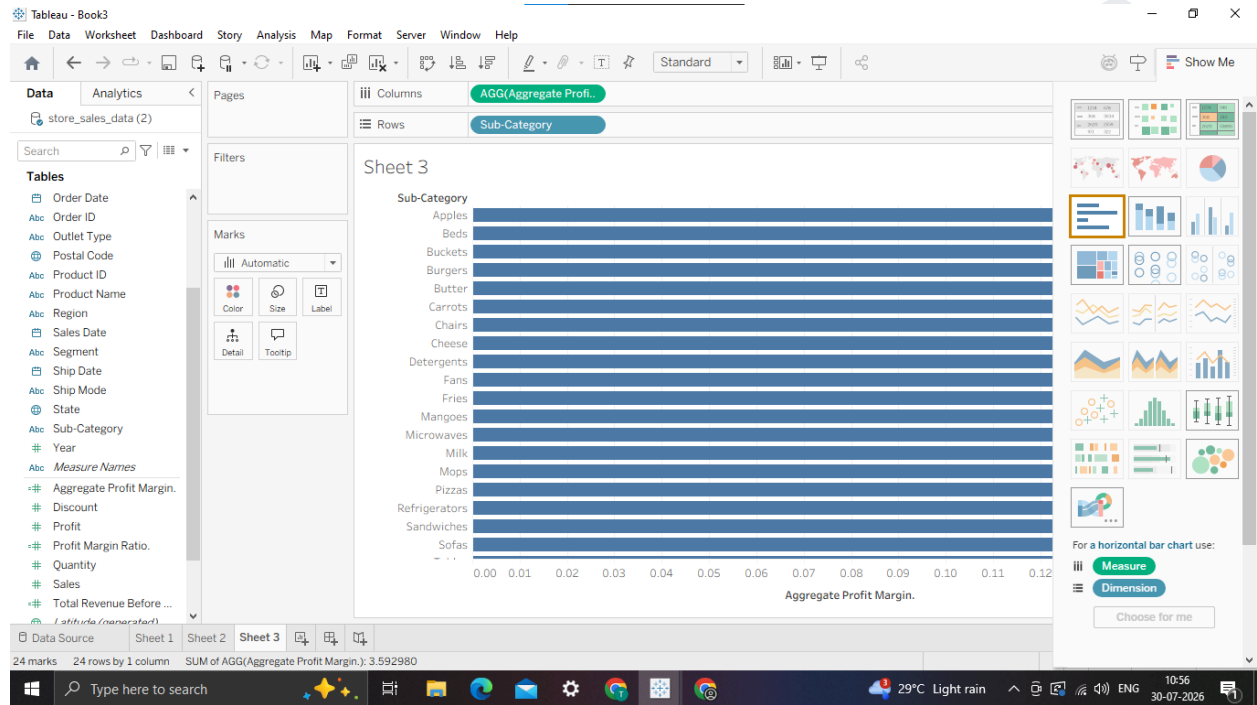
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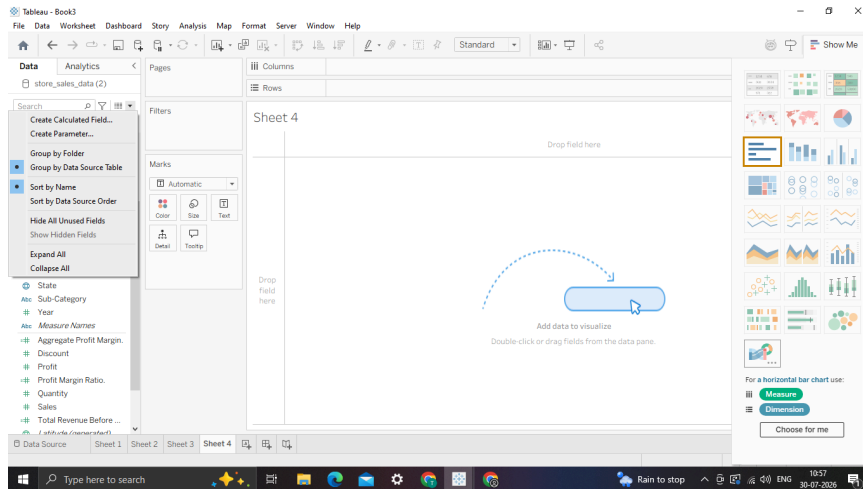
Profit Margin). Unlike standard fields, you cannot right-click this pill to change it to "Average" or "Count"—Tableau locks it as an aggregation because the math is already defined inside the formula.



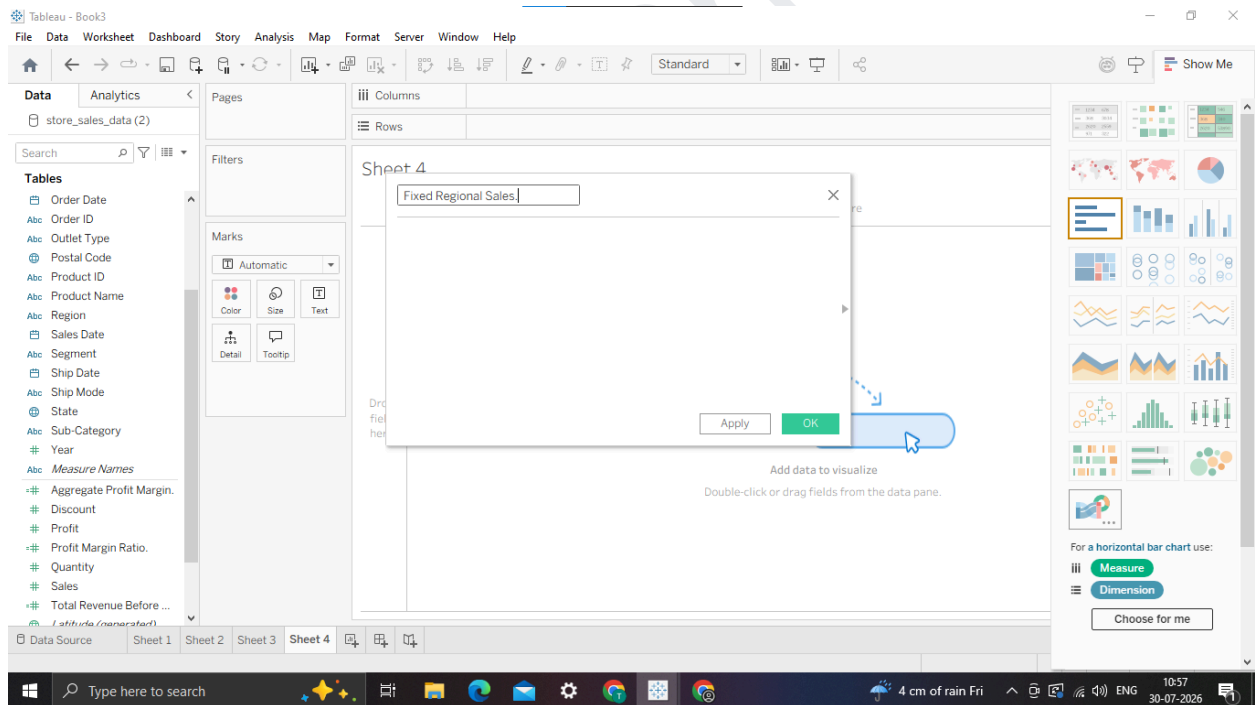
1. Creating a FIXED LOD Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...

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2. Name the field: Fixed Regional Sales.



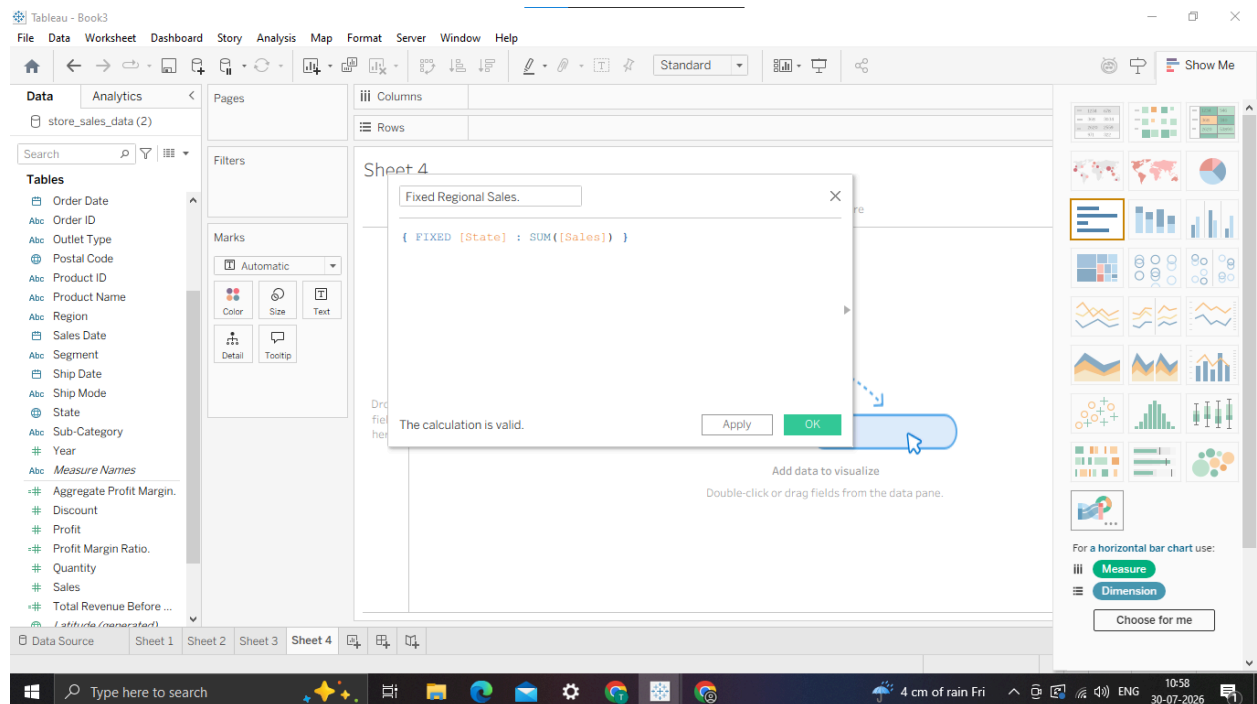
3. In the calculation formula window, type the following expression using curly braces

{:
{ FIXED [State] : SUM([Sales]) }

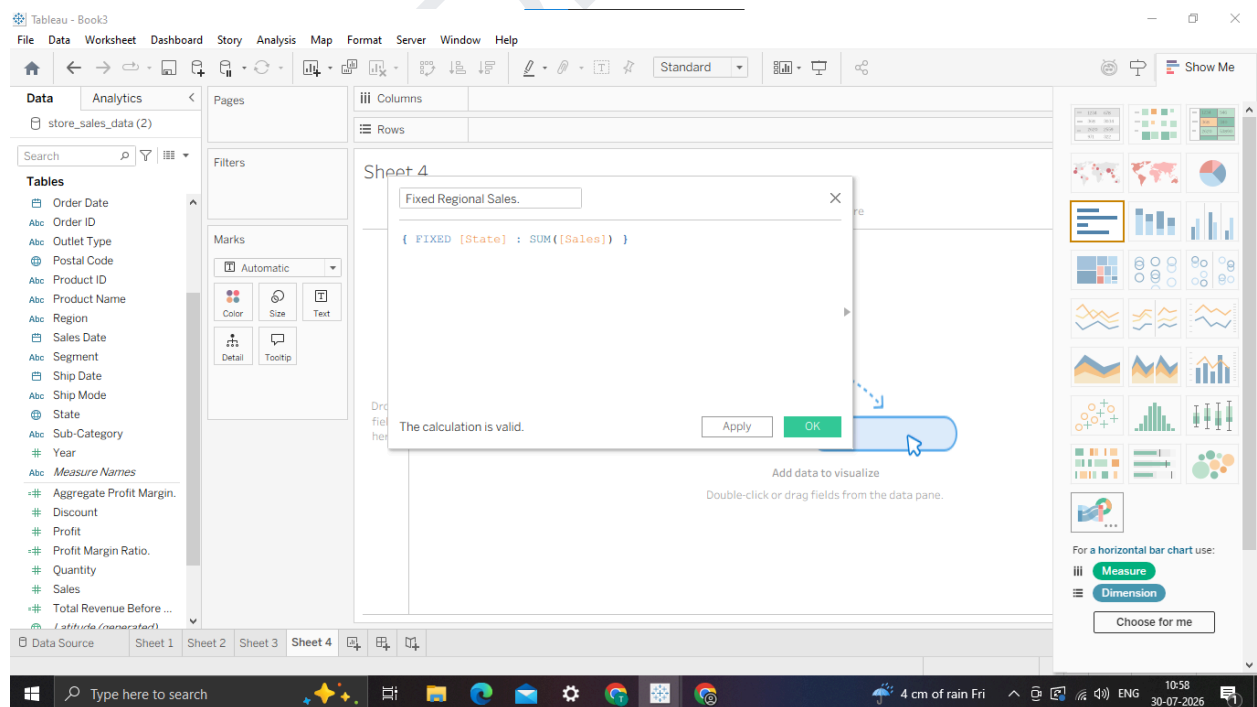
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4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



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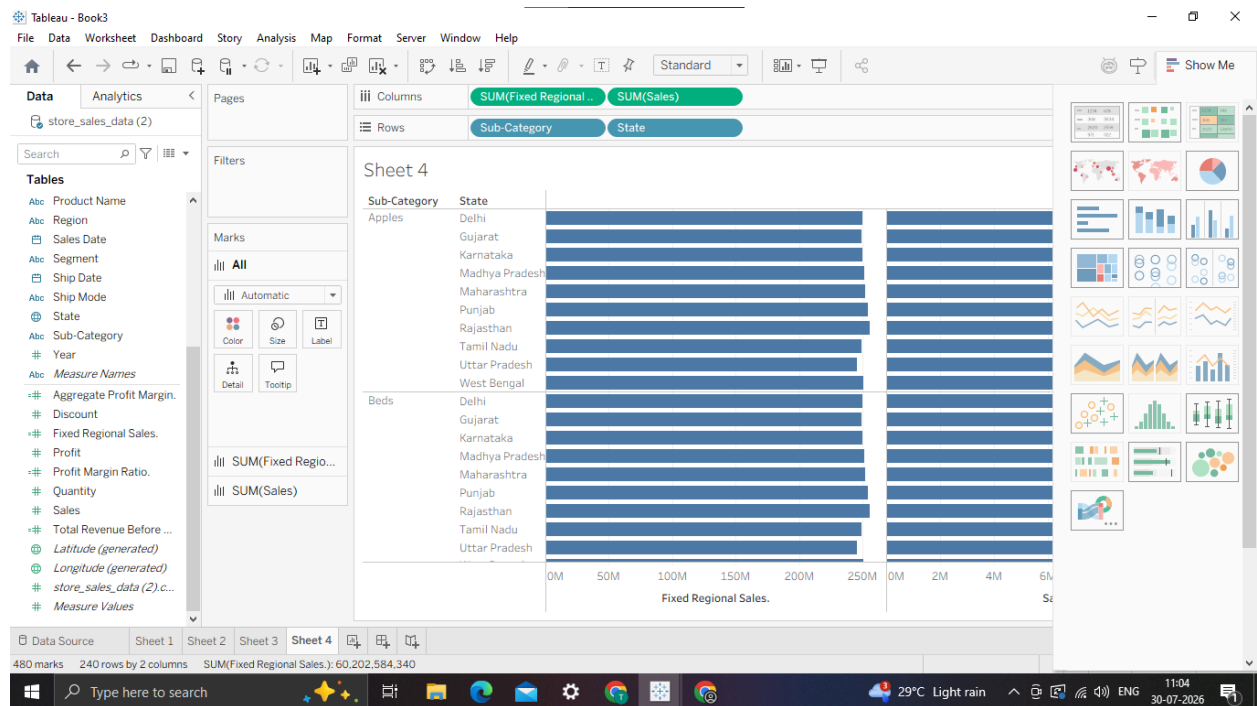
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2. Putting the Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State.
3. Drag the raw Sales measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state).
4. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.

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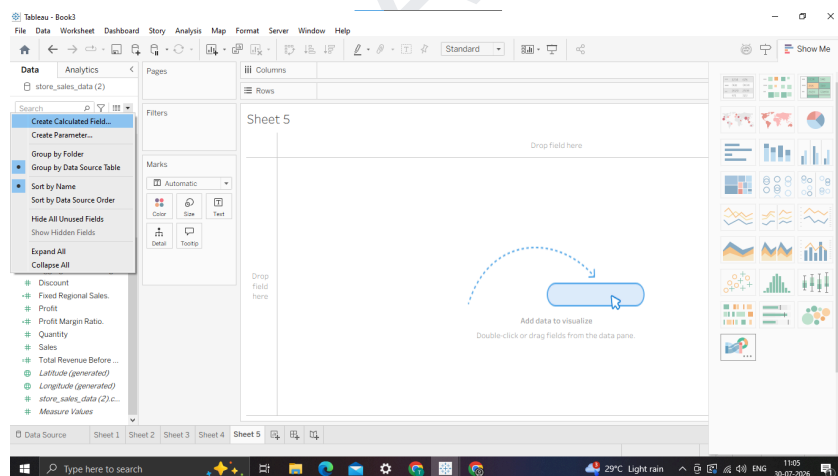
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E: Using INCLUDE and EXCLUDE LOD Expressions

1. Creating an INCLUDE LOD Calculated Field

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...

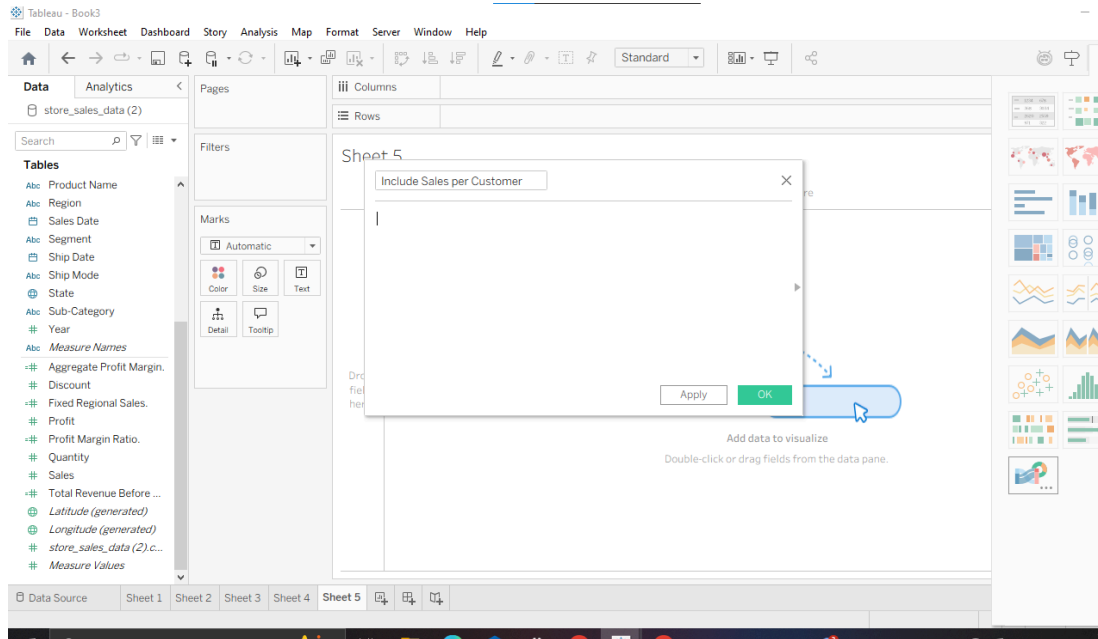


2. Name the field: Include Sales per Customer.

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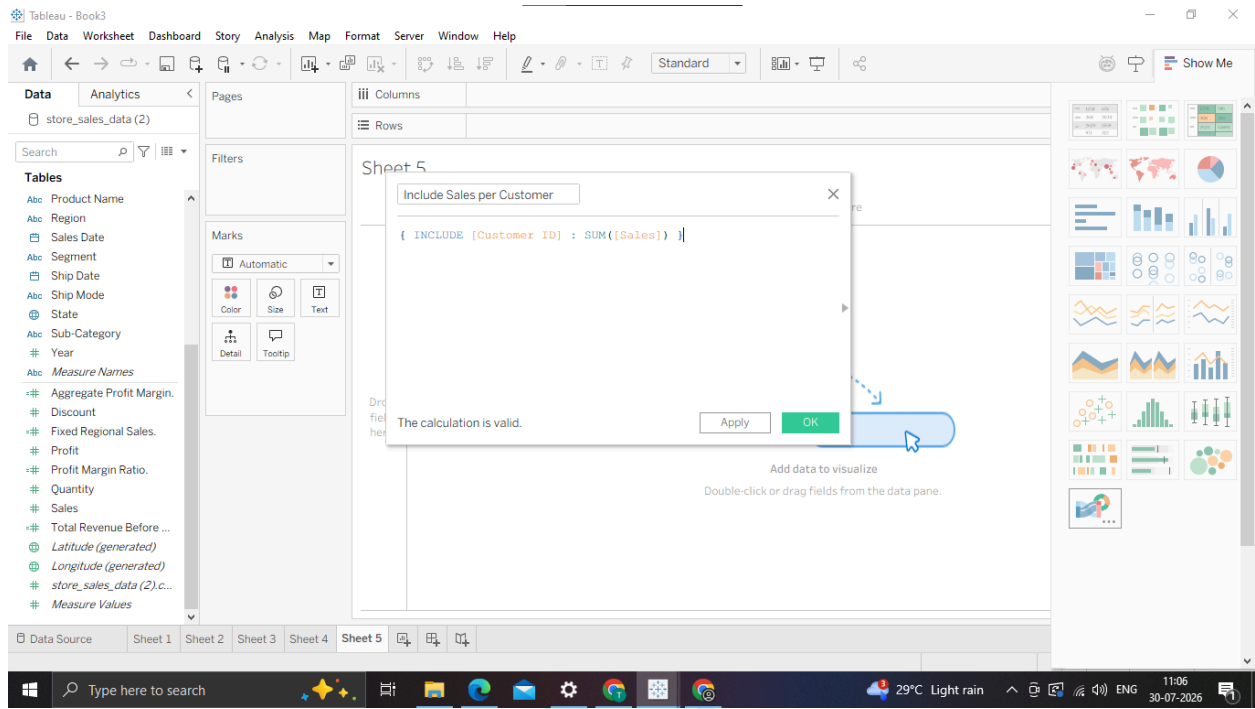
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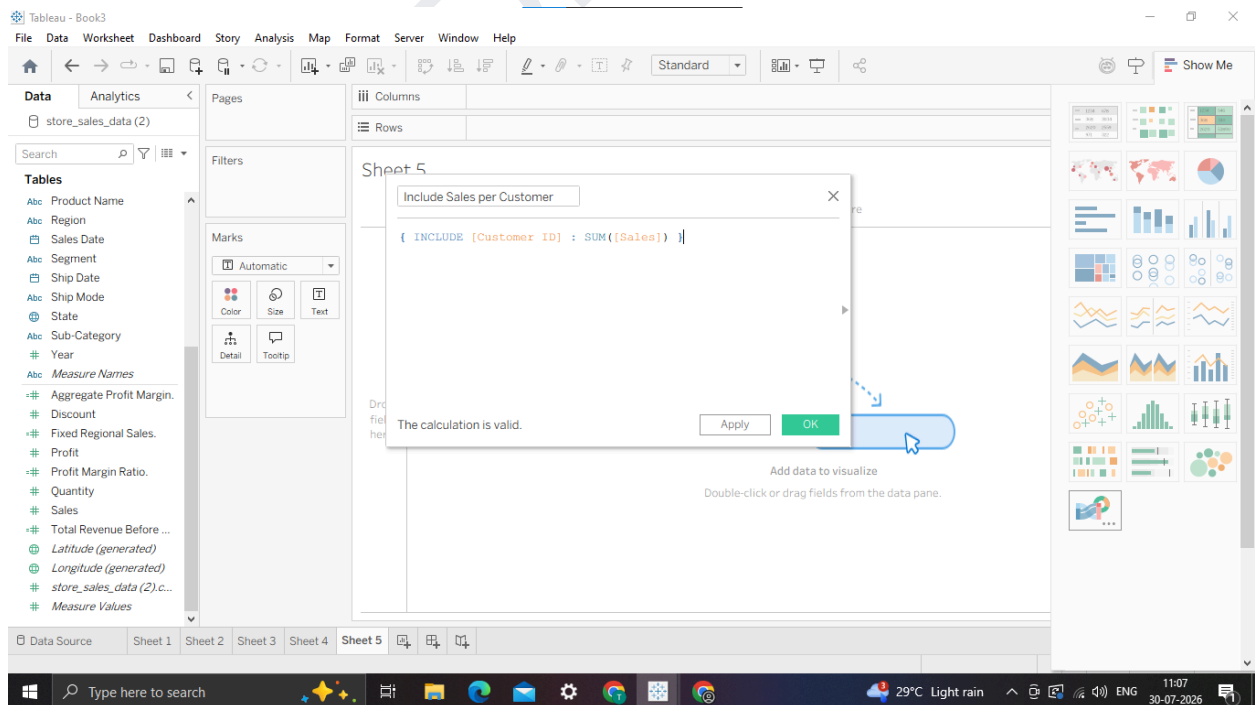


3. In the calculation formula canvas area, input the following expression:
{ INCLUDE [Customer ID] : SUM([Sales]) }

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4. Click OK.



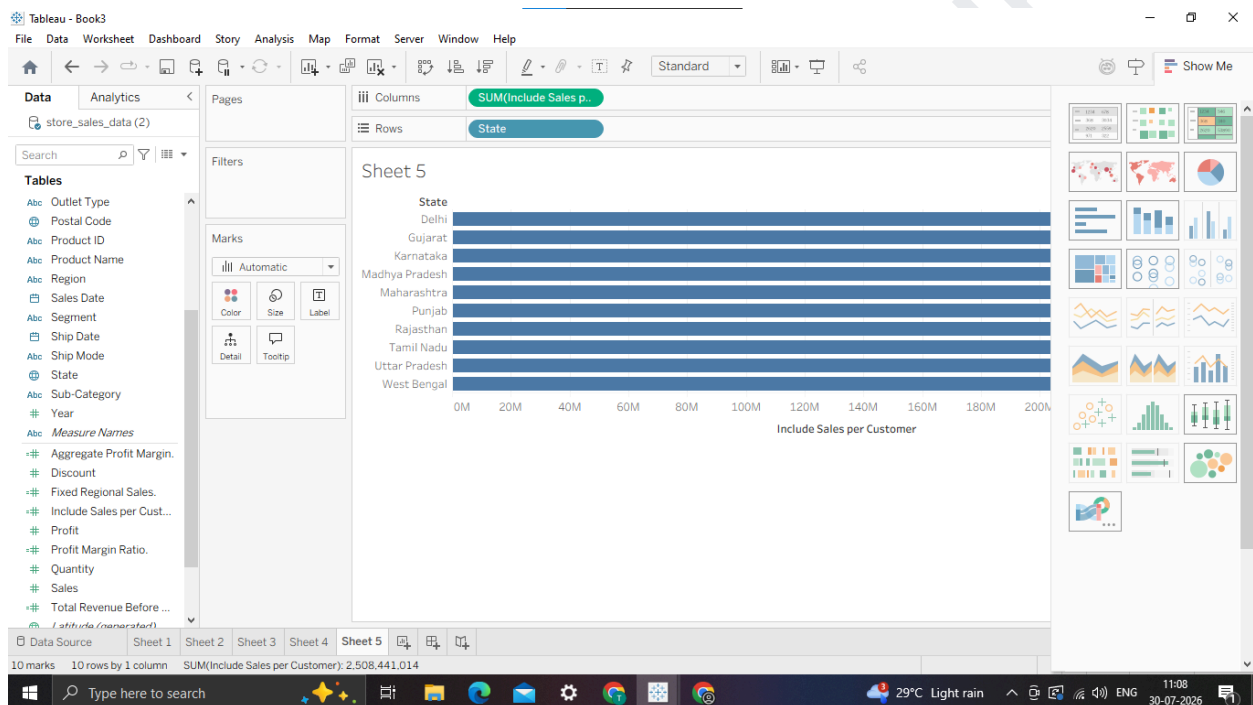
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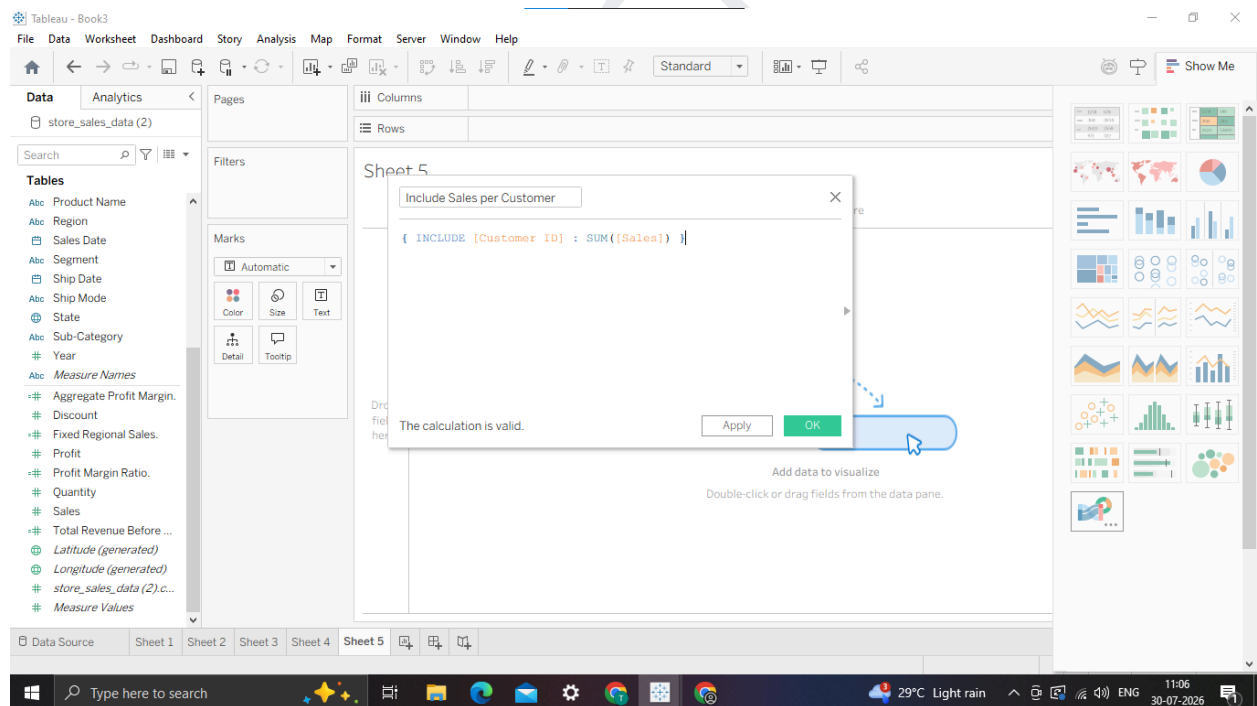
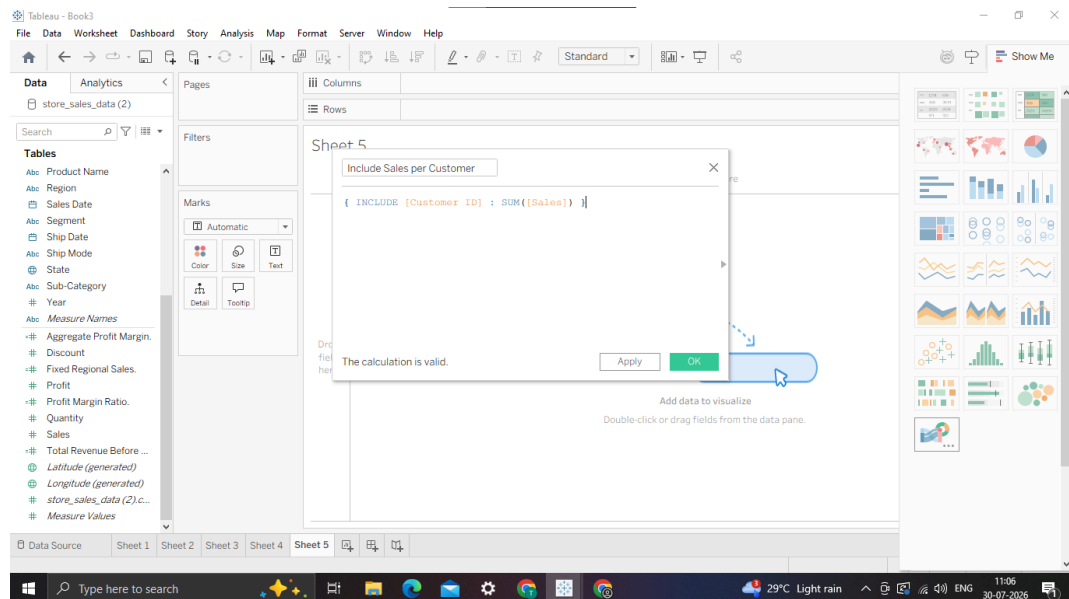
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2. Putting the INCLUDE Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Include Sales per Customer calculated field and drop it onto the Columns shelf.



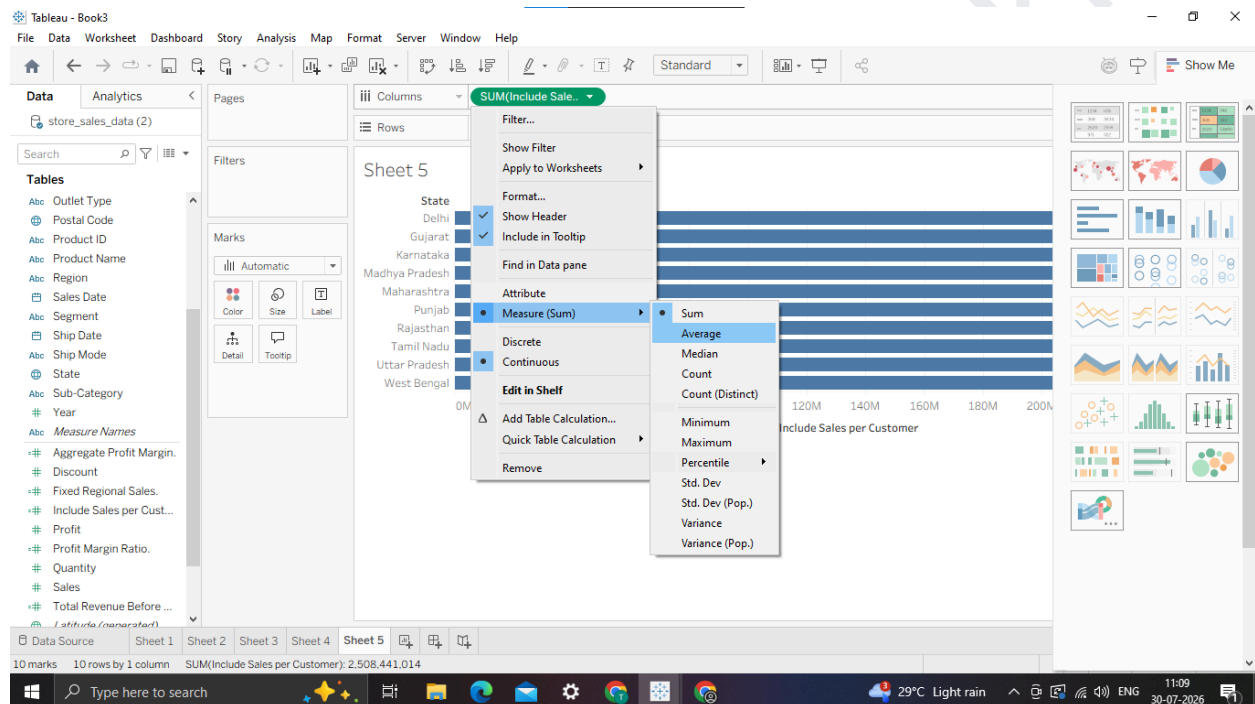
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3. Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.

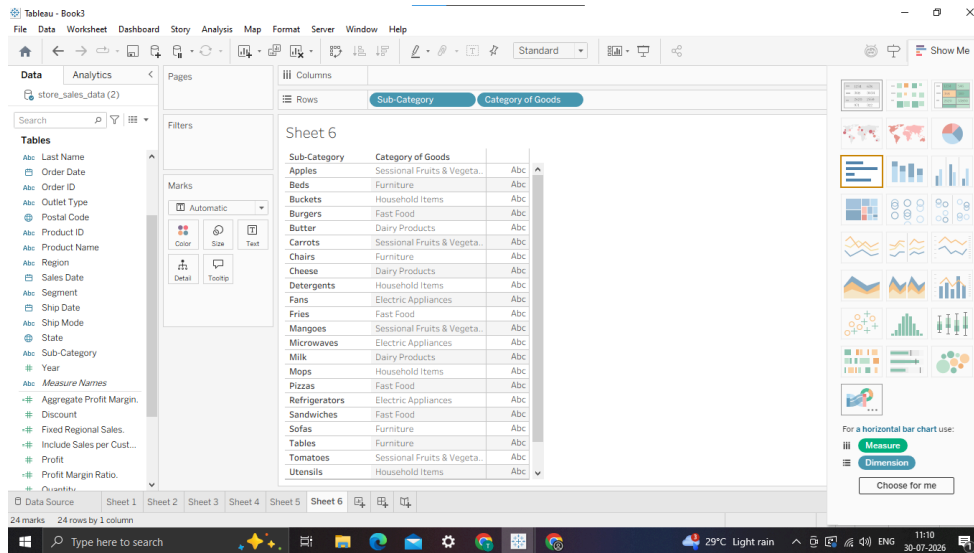


The screenshot displays the Tableau Desktop interface. The main view is a horizontal bar chart titled "Sheet 5". The chart's y-axis is labeled "State" and lists ten Indian states: Delhi, Gujarat, Karnataka, Madhya Pradesh, Maharashtra, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh, and West Bengal. The x-axis is labeled "Avg. Include Sales per Customer" and ranges from 0K to 20K in increments of 2K. The bars are blue. The interface includes a top menu bar (File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, Help), a left sidebar with "Data" and "Analytics" tabs, and a right sidebar with "Columns" and "Rows" shelves. The status bar at the bottom shows "10 marks 10 rows by 1 column SUM of AVG(Include Sales per Customer): 250,850.4".

3. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).

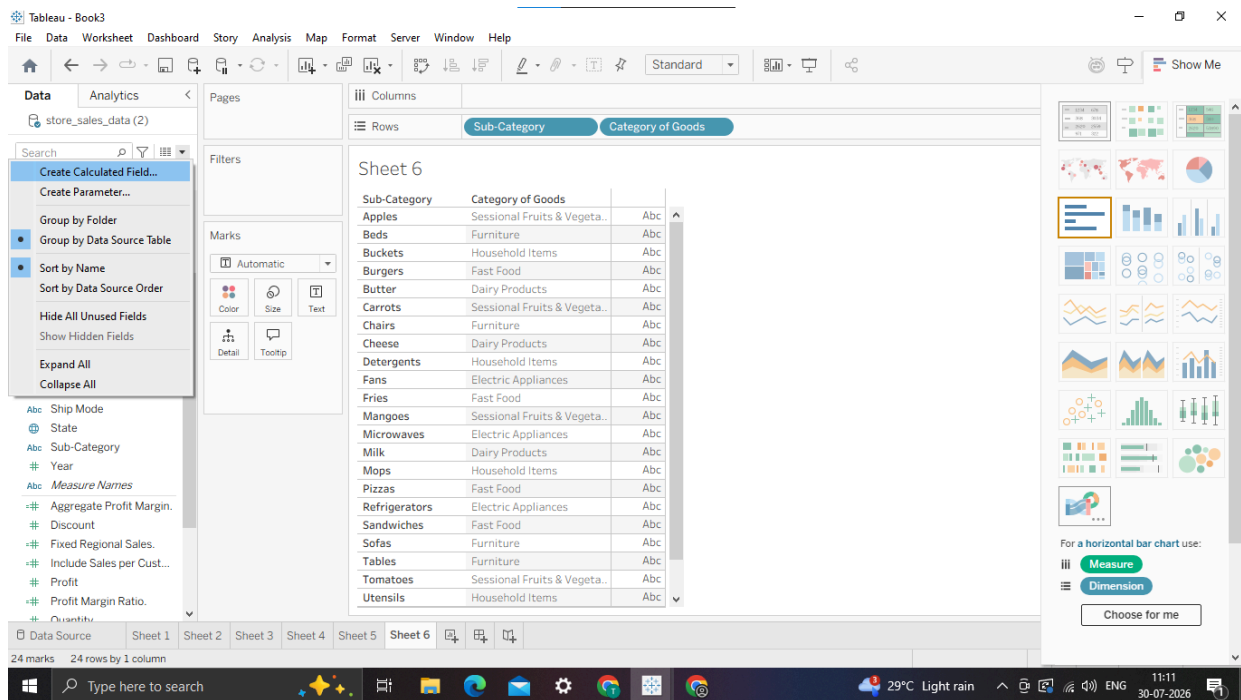
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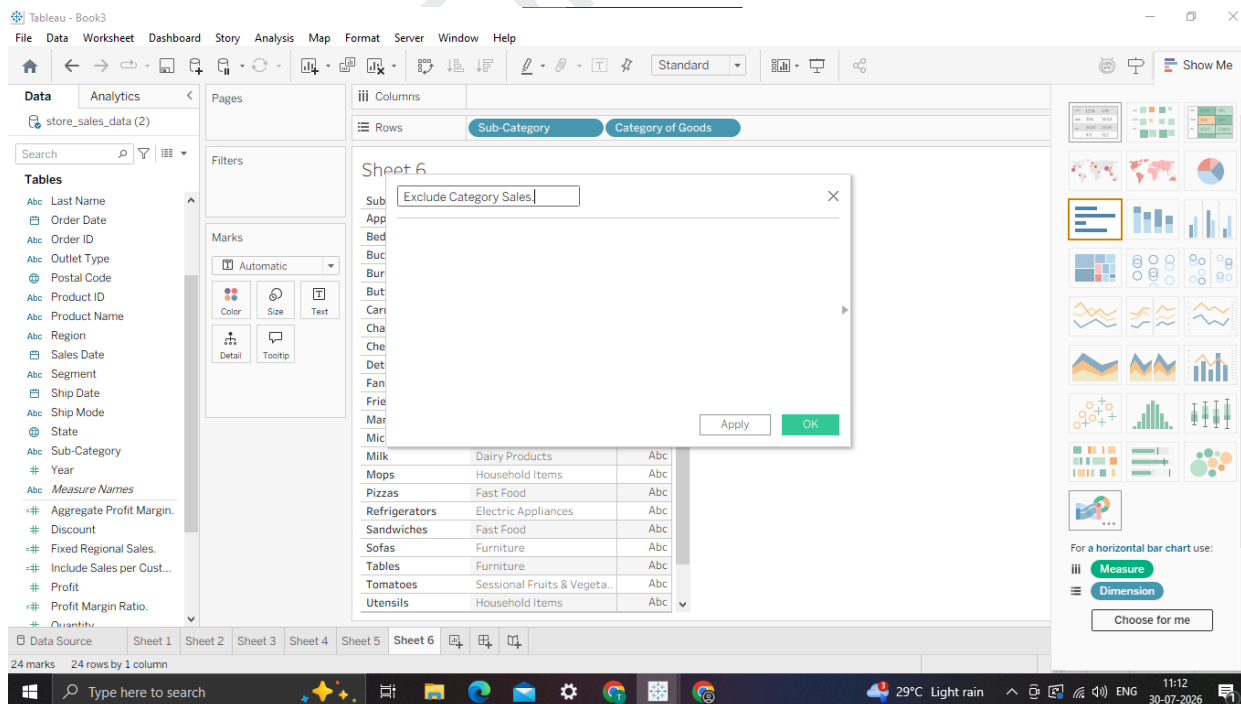


4. Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...

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5. Name the field: Exclude Category Sales.



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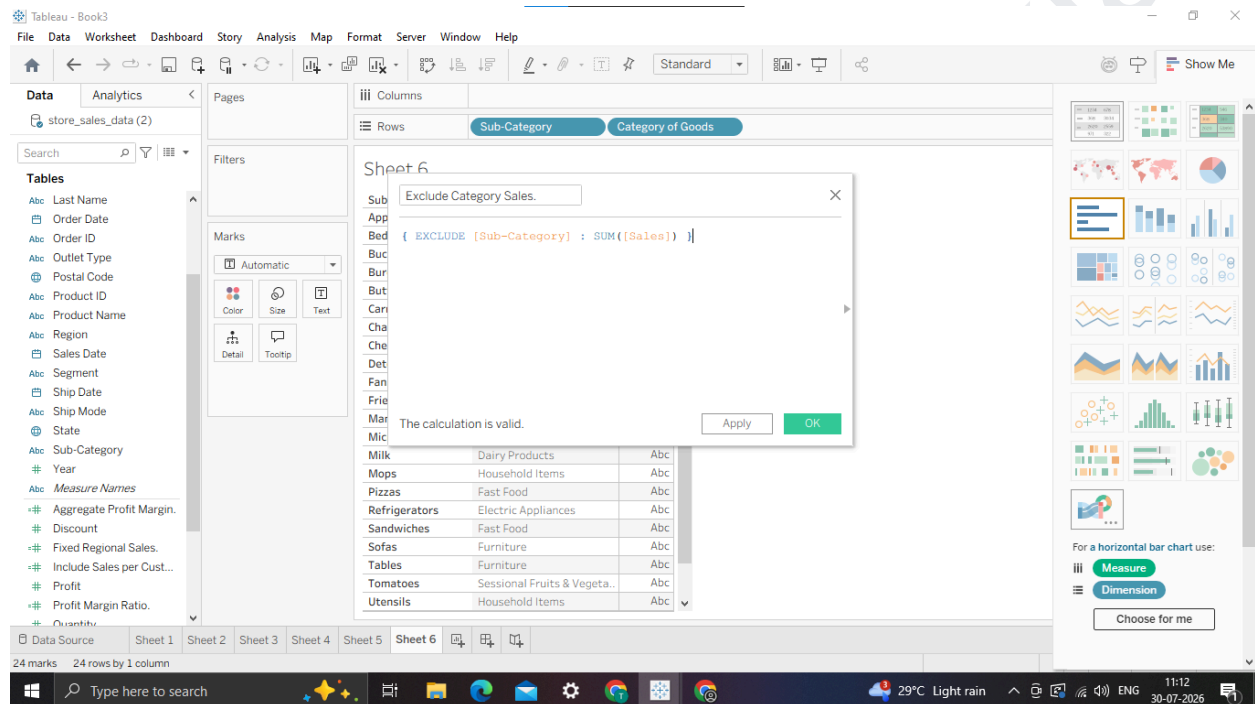
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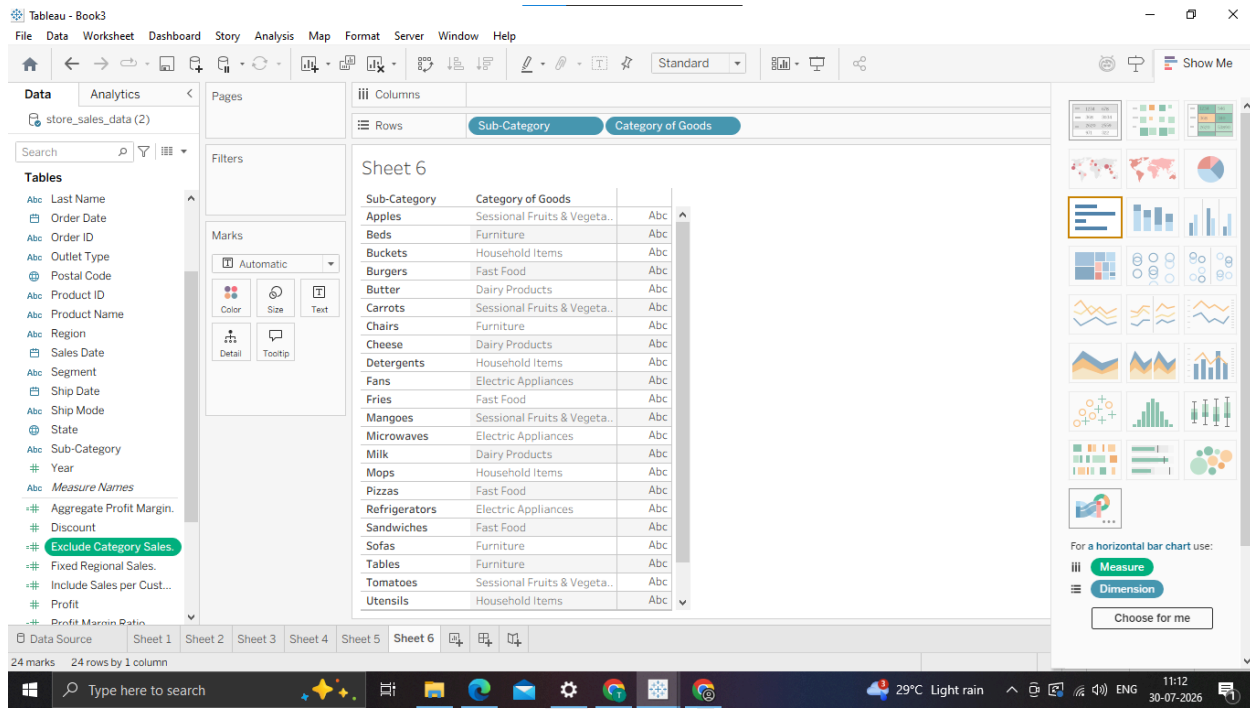
6. Input the following expression:

{ EXCLUDE [Sub-Category] : SUM([Sales]) }



7. Click OK and drag this new field onto the Columns shelf.

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F: Creating and Using Parameters for Dynamic Analysis

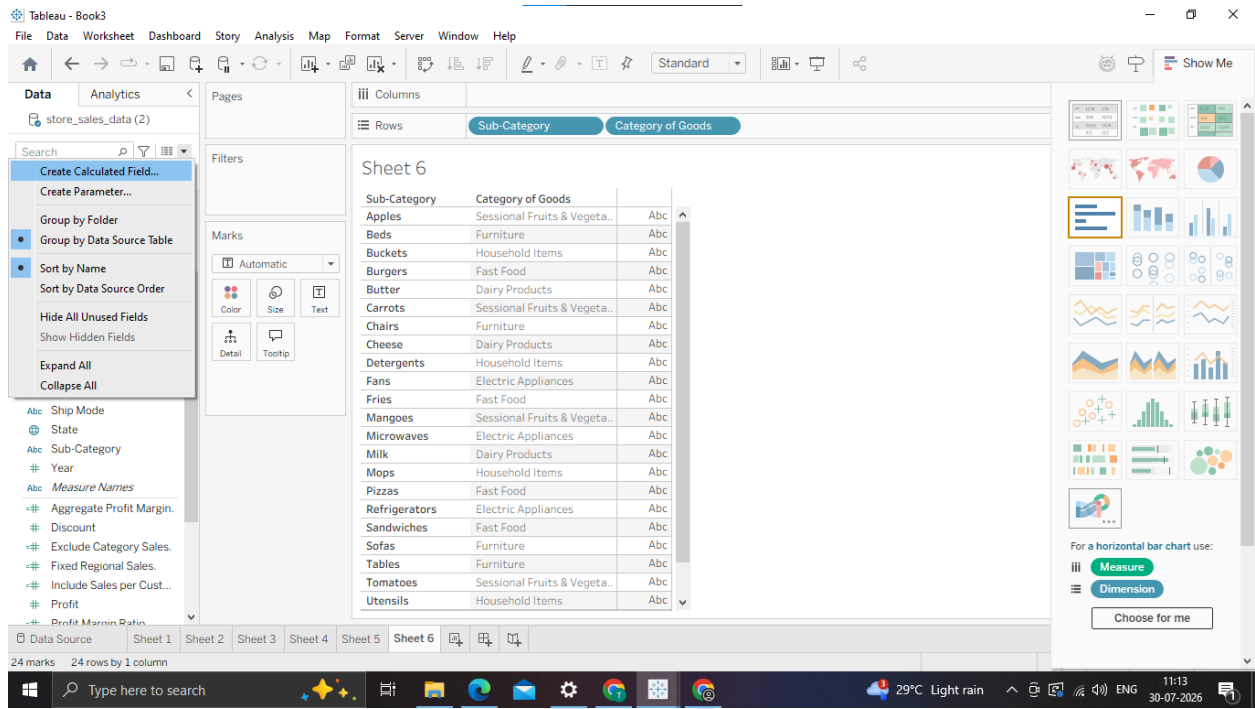
1. Creating a Parameter

Unlike calculated fields which evaluate data statically, a Parameter is a dynamic variable or workbook input control block (like a slider or dropdown) that allows users to alter calculations or values on the fly without changing the underlying architecture.

Let's build a parameter that lets users set their own dynamic Profit Target threshold:

1. Go to your Data Pane. Click the small drop-down arrow (small triangle) next to the search bar at the top and choose Create Parameter.

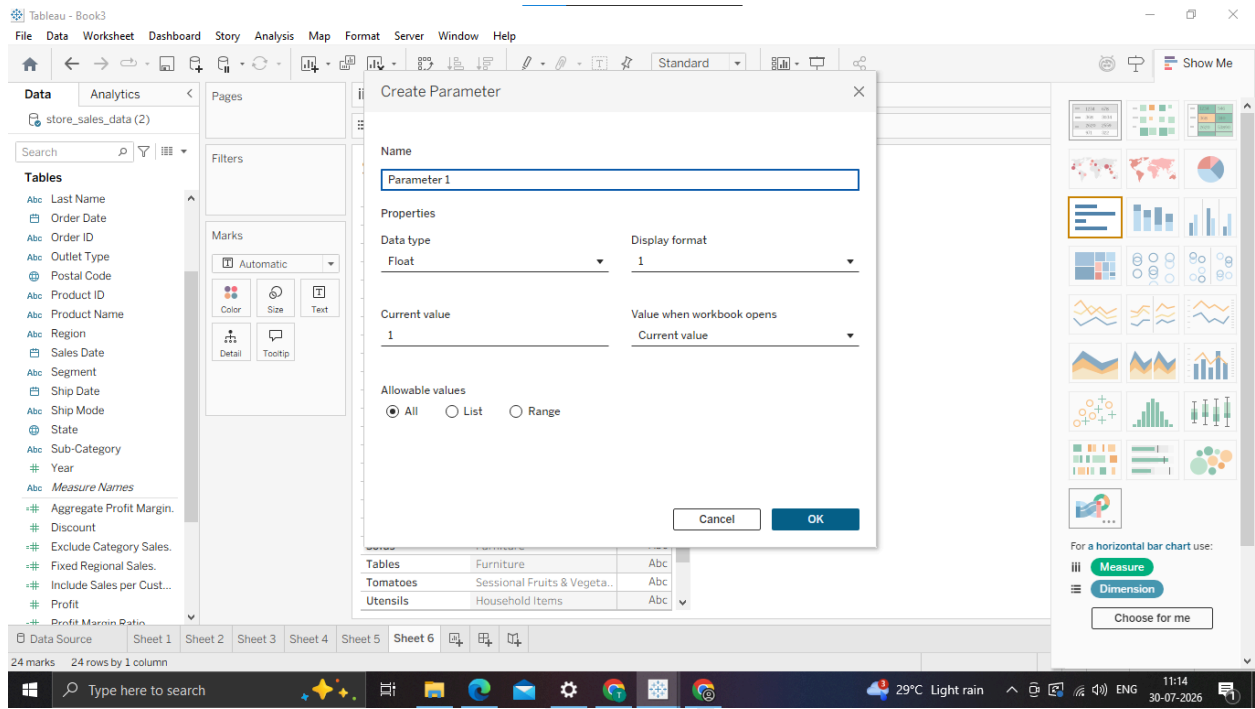
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2. Name the parameter: Select Profit Threshold.

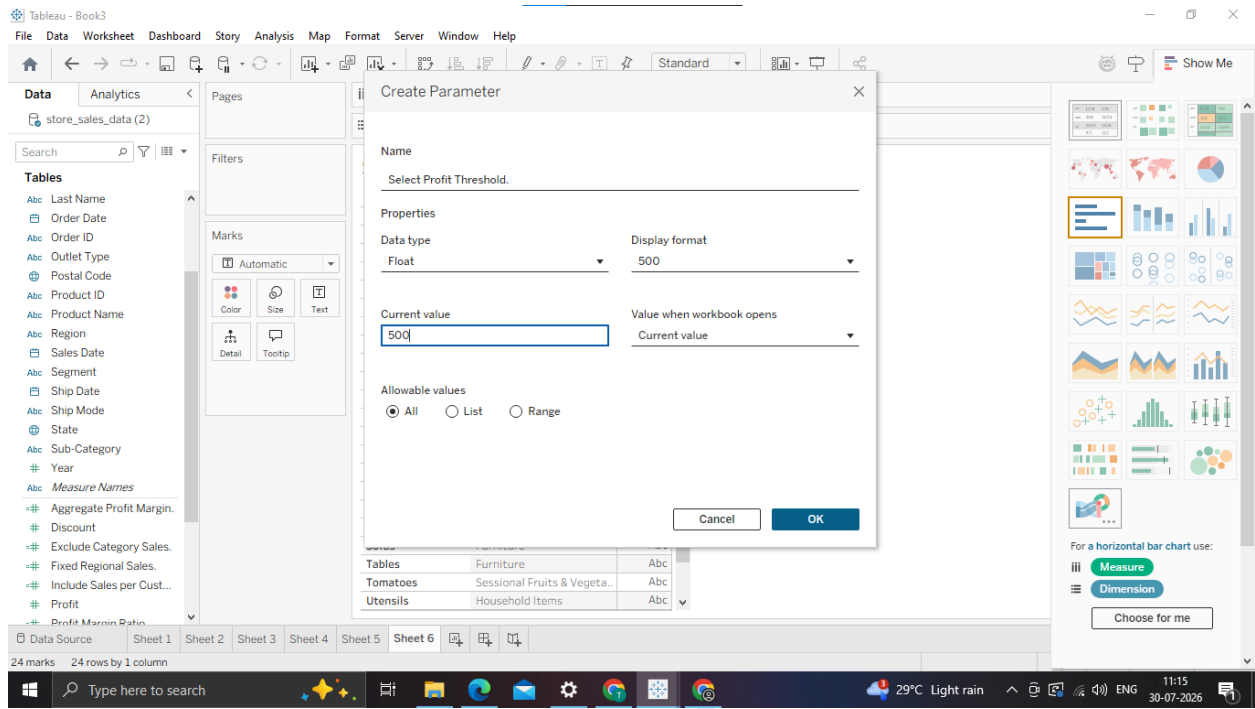
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3. Set the Data Type to Float (or Integer).

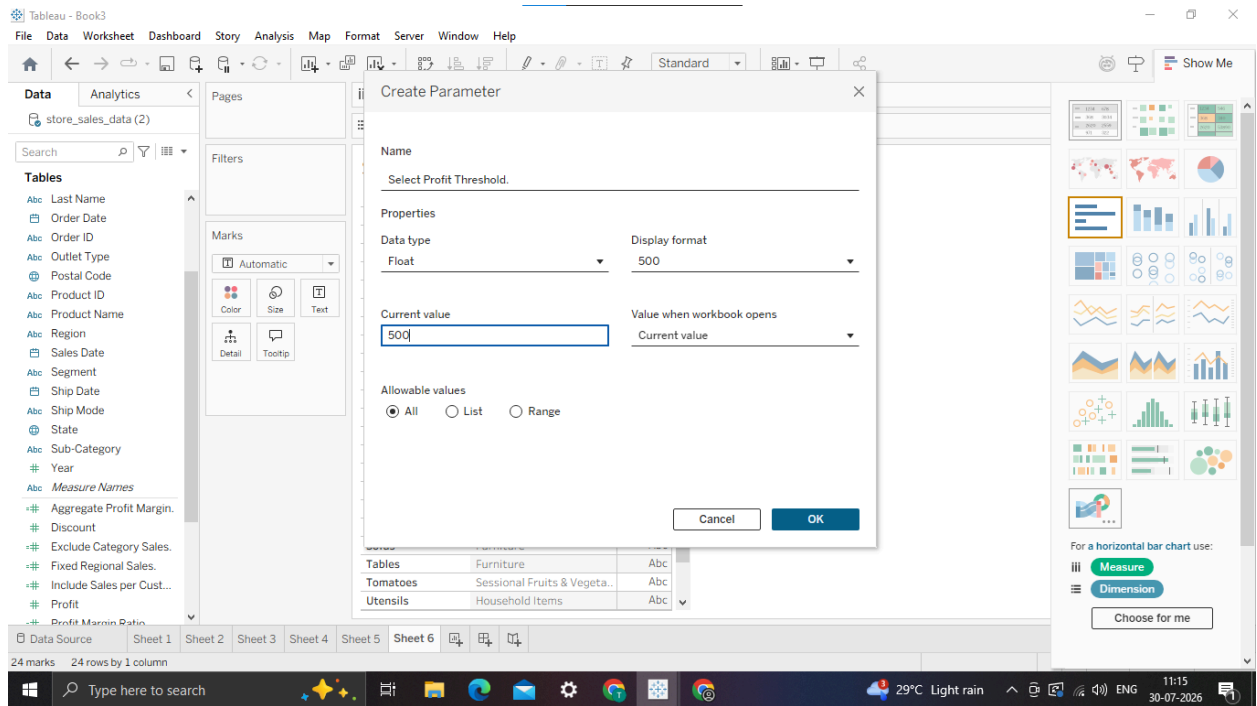
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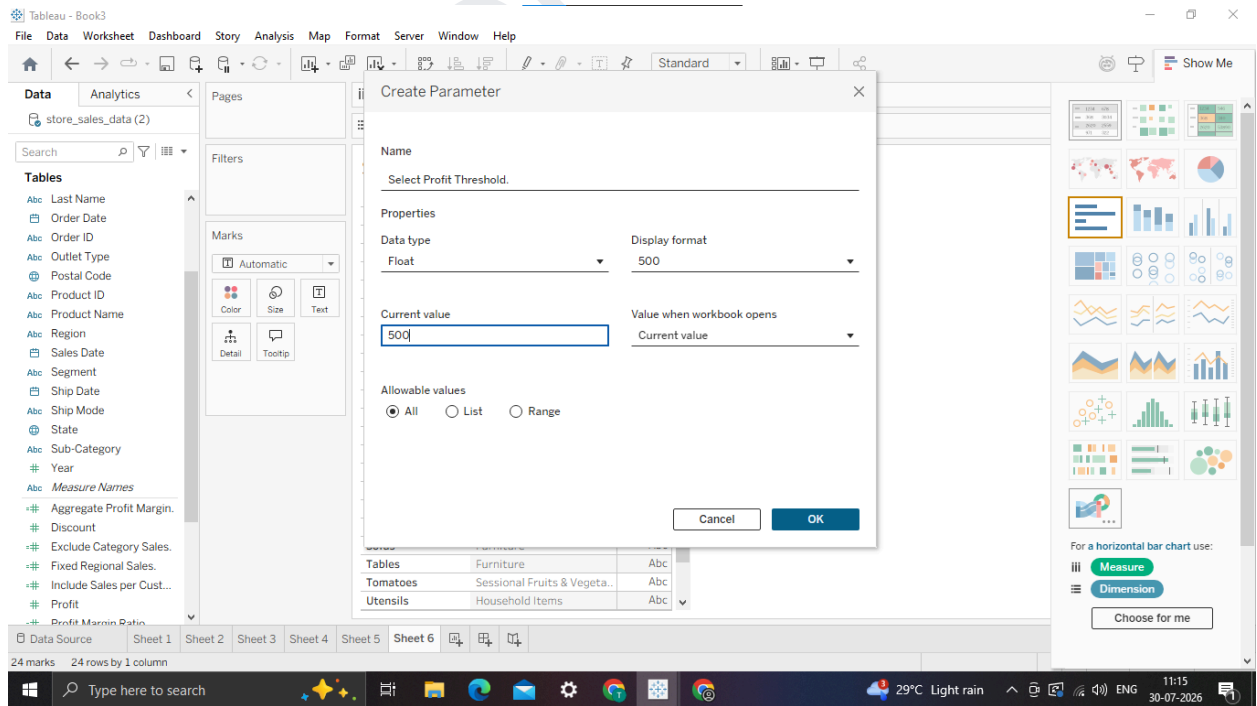
4. Set the Current Value to 500.

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5. Keep the Allowable values set to All and click OK.

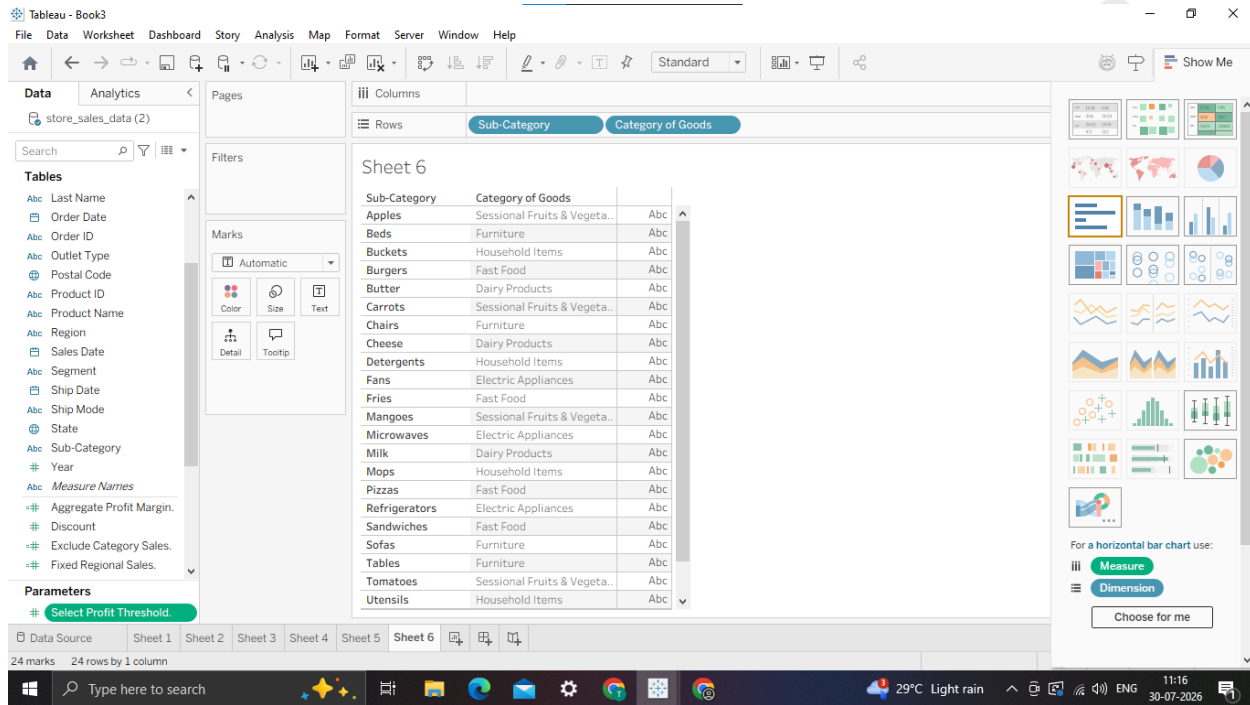


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6. Notice a new section titled Parameters appears at the bottom of your Data Pane, showcasing your new selector.

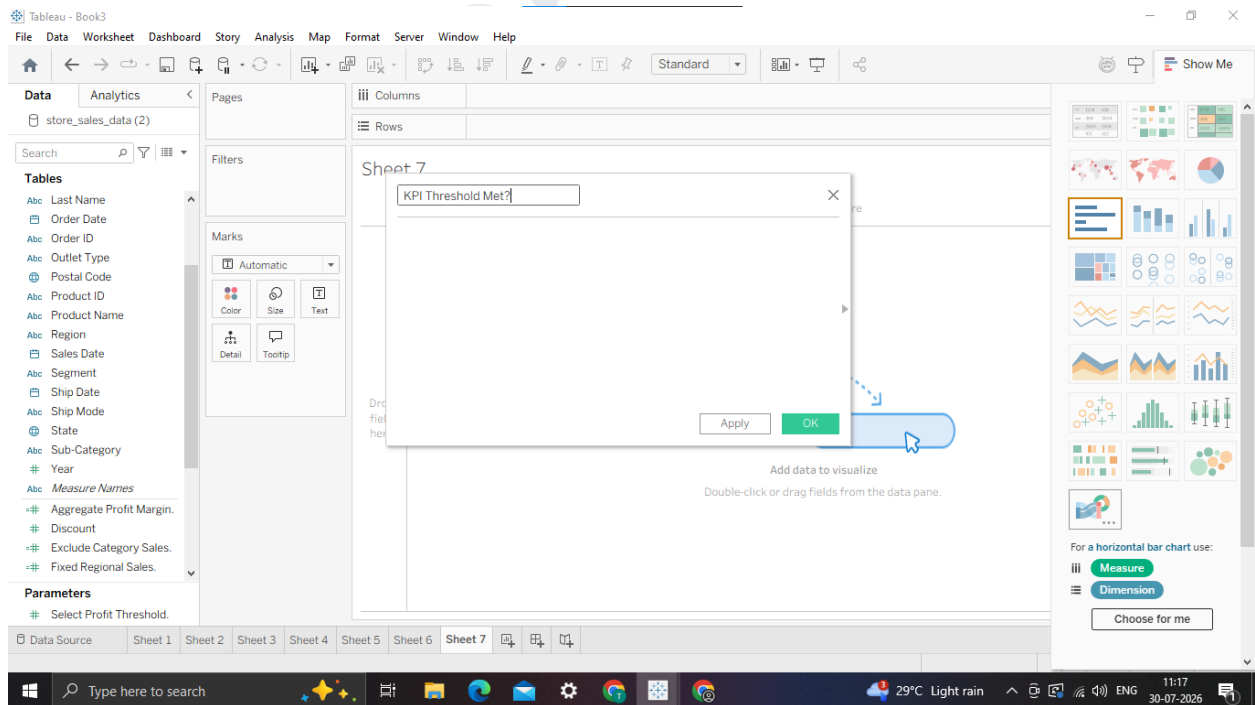
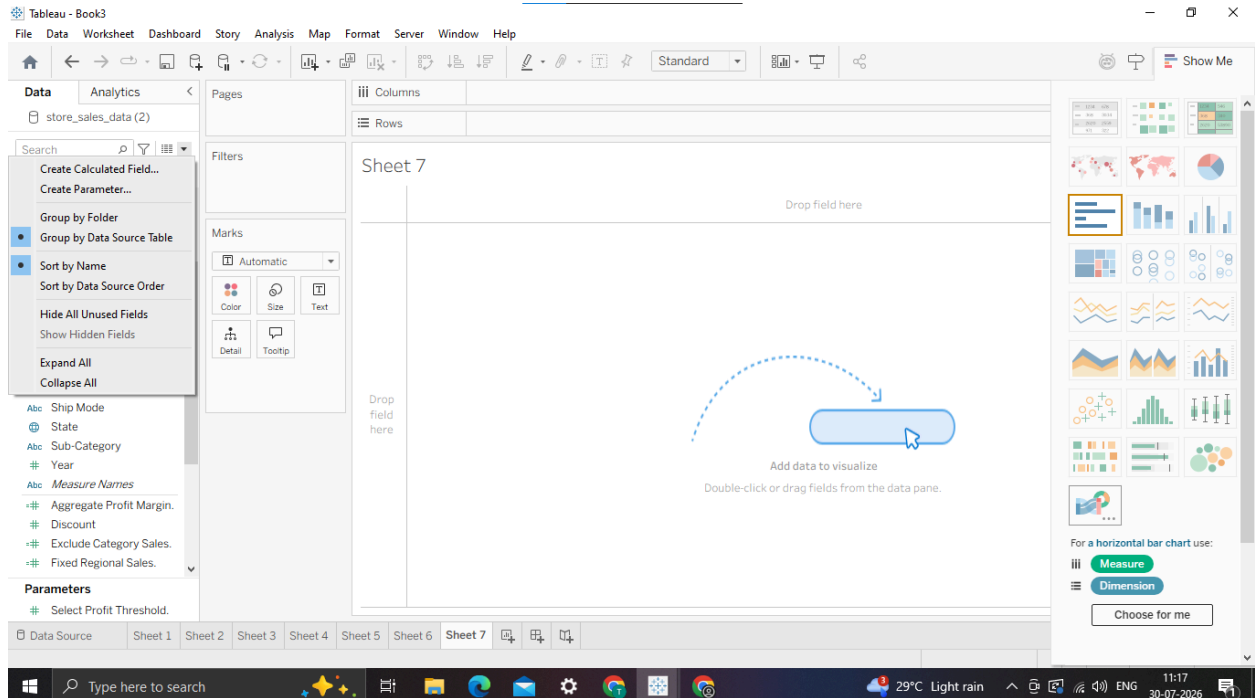


2. Linking the Parameter to a Calculated Field

A parameter does absolutely nothing until you actively tie it to a field layout or formula. Let's create a conditional formula that flags categories based on the parameter's current value:

1. Right-click in the blank white space of the Data Pane or Click on small triangle angle next to search bar and select Create Calculated Field...
2. Name this field: KPI Threshold Met?.

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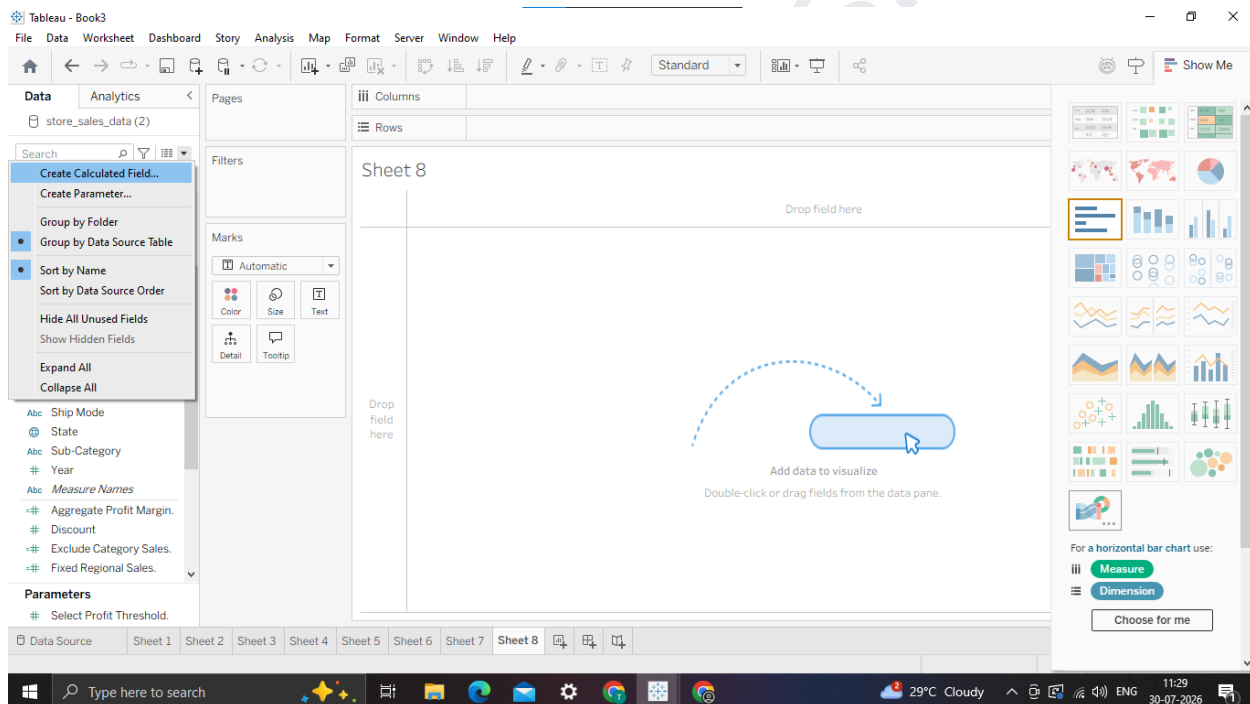
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1. Handling Null or Missing Values with IFNULL

In enterprise datasets, missing data or null entries can cause calculations to break or display empty spaces in charts. We can use a logical calculation to substitute a clean value whenever a null occurs.

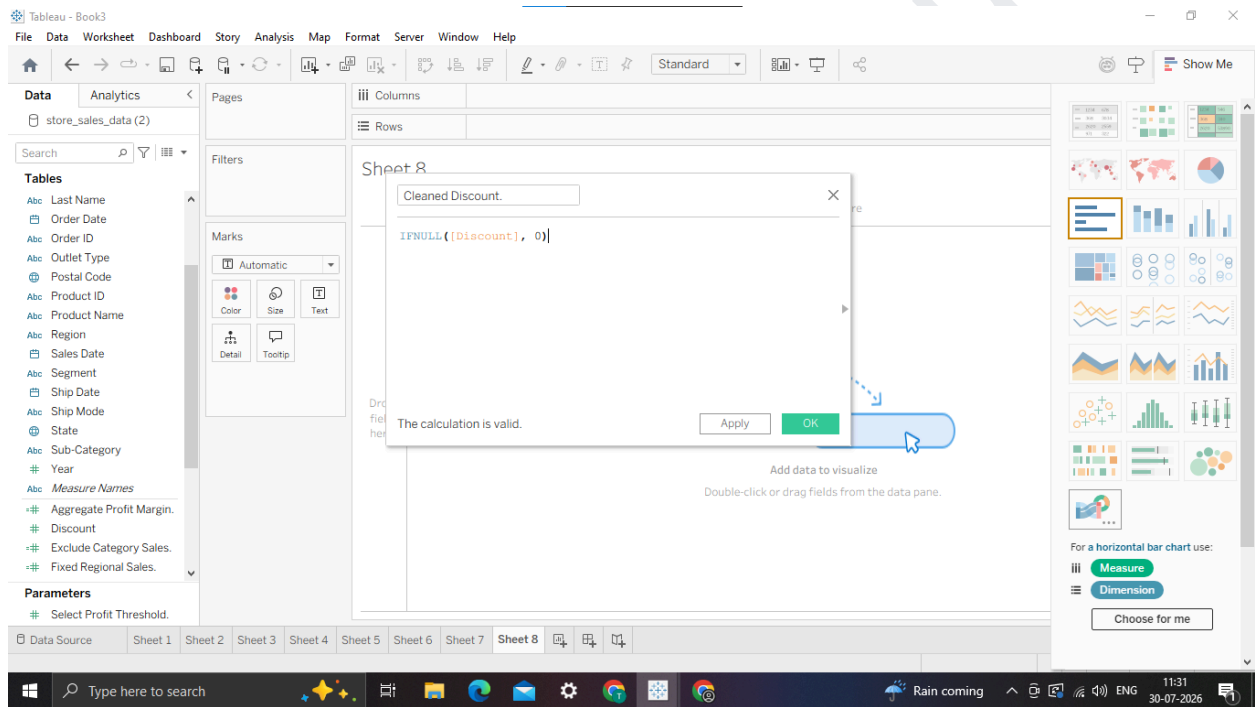
Let's clean up potential missing indicators in our data:

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...

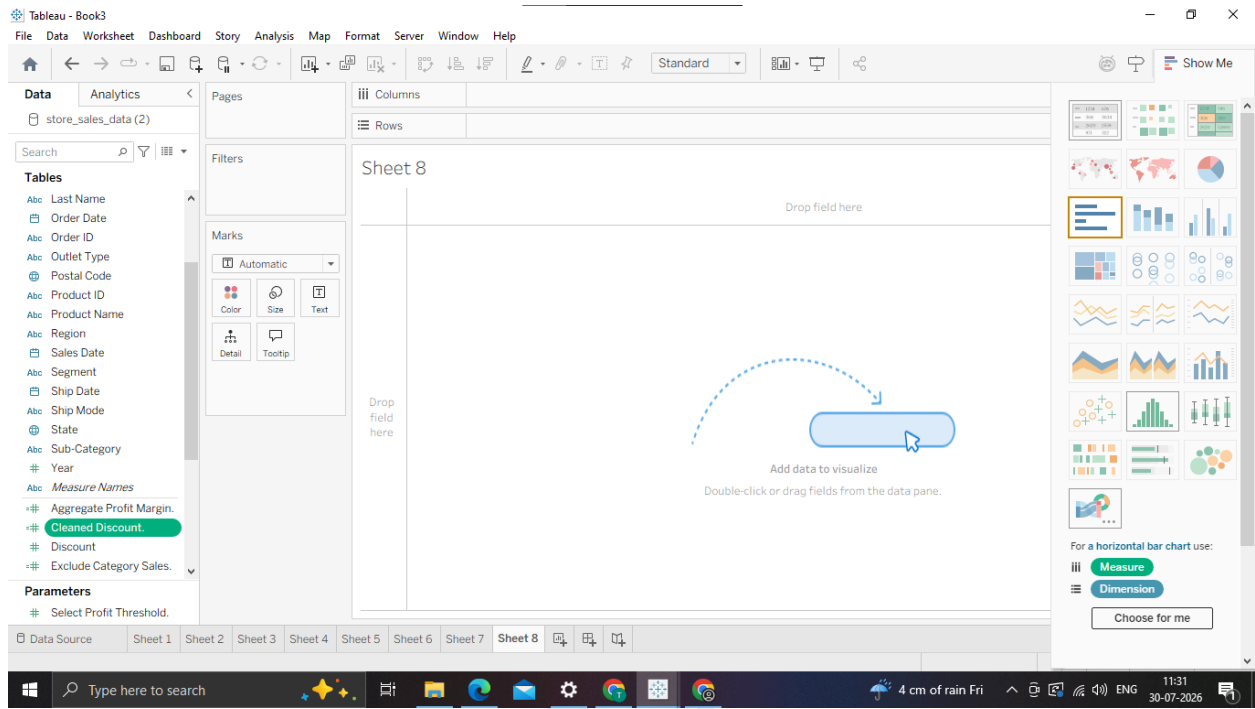


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2. Name the field: Cleaned Discount. 3. Input the following expression:
IFNULL([Discount], 0)



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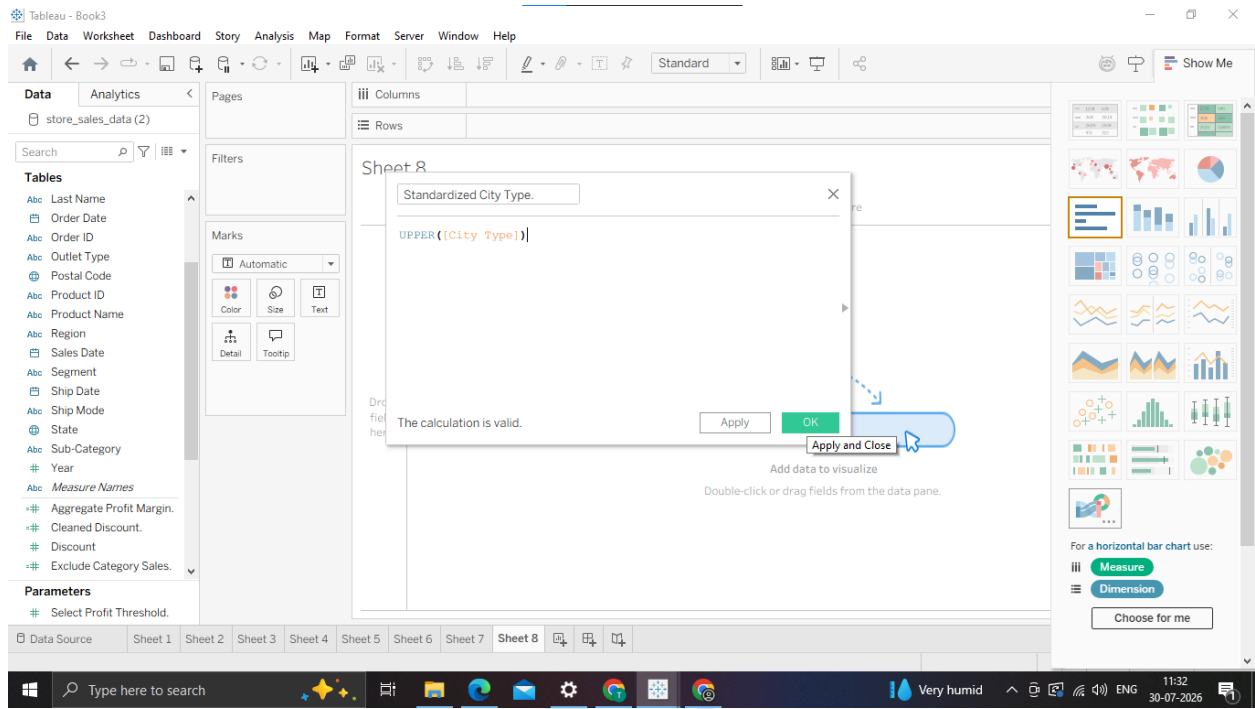


2. Segmenting Data for Insights using String Functions (UPPER/LOWER)

Sometimes database values are entered with inconsistent casing (e.g., "tier 1", "Tier 1", "TIER 1"), which causes Tableau to split them into separate visual blocks. We can force uniform compliance using a string calculation:

1. Create a new Calculated Field named Standardized City Type.

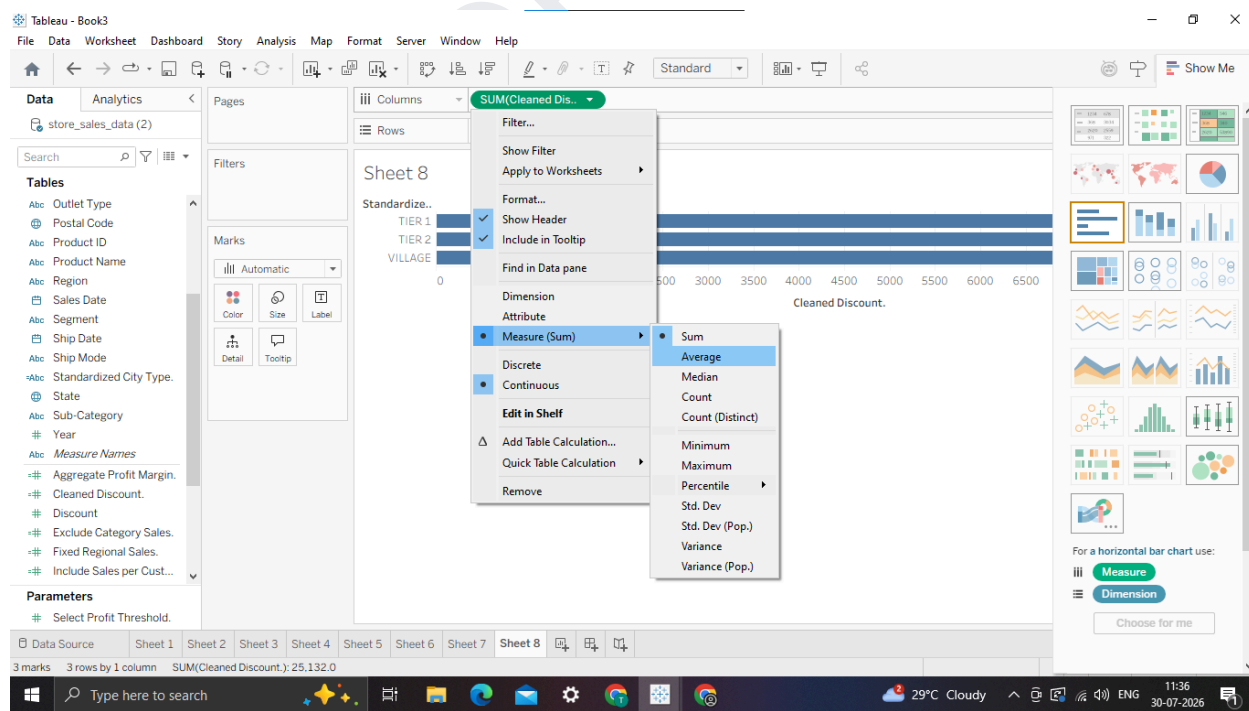
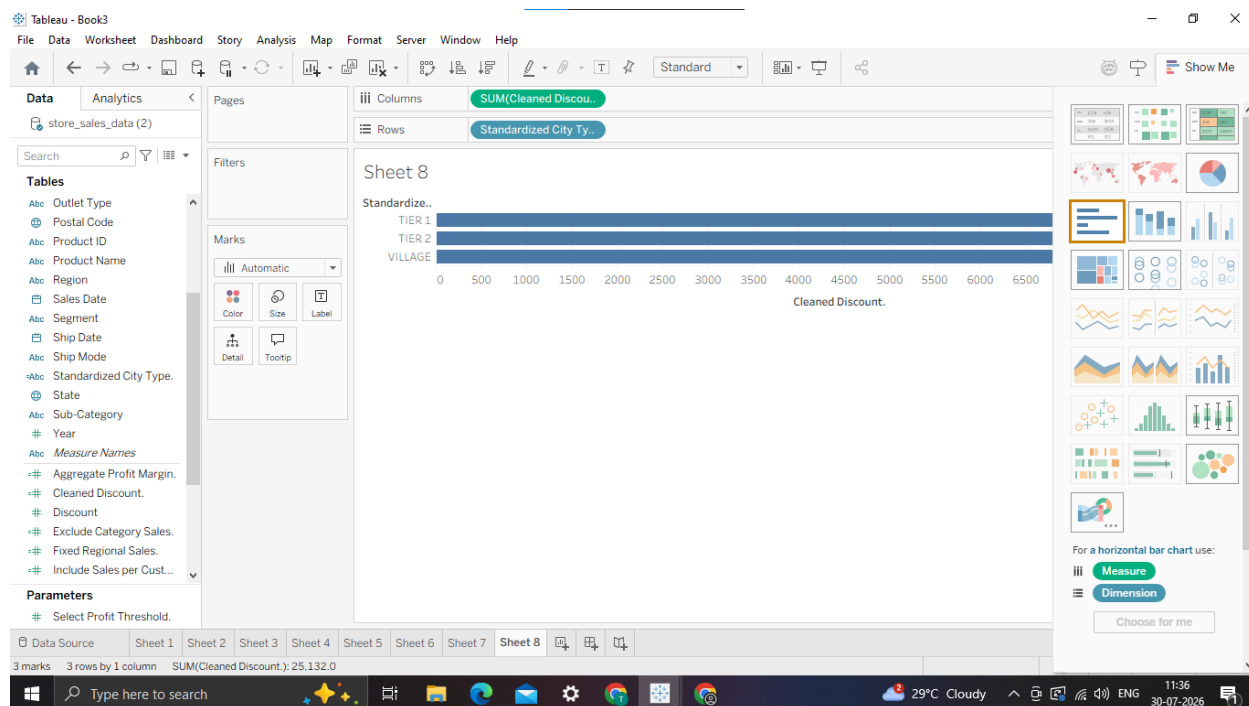
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3. Putting the Cleaned Fields to Use

1. Open a new blank worksheet.
2. Drag your newly created Standardized City Type dimension from the Data Pane and drop it onto the Rows shelf.

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